Accounting for the Urban Wage Growth Premium

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Abstract

Why do wages grow faster in bigger cities? We use French administrative data to decompose the urban wage growth premium and find that the answer has surprisingly little to do with cities themselves. While we document substantially faster wage growth in larger cities, 57% of the premium is accounted for by the composition of firms and coworkers. We also document significantly higher job-to-job transition rates in larger cities, suggesting workers climb the job ladder faster. Most strikingly, when we focus on workers who remain in the same job, eliminating the job ladder mechanism, the premium falls 84% below the full-sample baseline once firms and coworkers are accounted for. These results challenge the view that cities generate human capital spillovers "in the air," suggesting instead that urban wage dynamics reflect the sorting of firms and workers and the pace of job mobility.

JEL codes: R3, J2

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Introduction

We have known, at least since Smith (1776), that the returns to labor differ across the city size distribution, and that these differences are deeply intertwined with the sorting of workers across space:

> *Industry, therefore, must be better rewarded, the wages of labour and the profits of stock must evidently be greater, in the one situation than in the other [town vs. country]. But stock and labour naturally seek the most advantageous employment. They naturally, therefore, resort as much as they can to the town, and desert the country.*

One of the central tasks of urban and regional economics has been to understand why wages are higher in bigger cities. Does it reflect spillovers that become stronger in dense cities, or does it reflect sorting of firms and workers? The answer to this question has important implications for the design of place-based policies and land-use regulations.

Famously, Marshall (1890) argued that *"the mysteries of the trade become no mysteries; but are as it were in the air"*, implying ambient learning spillovers from neighbors. This has become somewhat of a canonical explanation for why wages grow faster in bigger cities, i.e. the idea that city size, in and of itself, generates spillovers for all workers in a location. In this paper, we revisit the urban wage growth premium, the faster growth of wages in bigger cities (as distinct from the static premium in wage levels), and provide new evidence that rather than reflecting these ambient spillovers, it is largely accounted for by the firms and coworkers that sort into cities, with the remainder concentrated among workers who change jobs in cities where job changes are more frequent.

In particular, we are motivated by a burgeoning literature on the role of coworkers in within-firm human capital accumulation (Nix, 2020, Jarosch, Oberfield, and Rossi-Hansberg, 2021, Gregory, 2023, Herkenhoff, Lise, Menzio, and Phillips, 2024, Adenbaum, Babalievsky, and Jungerman, 2024) which has found that a substantial fraction of on-the-job learning is attributed to having more-skilled coworkers. If bigger cities have better firms and more skilled coworkers, could the urban wage growth premium reflect who workers work with, and who they work for, rather than *where* they work? In this paper, we ask this question directly. Our results suggest a reinter-

pretation of the dynamic premium documented in the seminal works of Glaeser and Maré (2001) and De la Roca and Puga (2017), which were estimated on data that do not observe the firm and its coworkers.

We focus on France and use rich matched employer-employee administrative data that allow us to track workers' labor market trajectories over time to first document that the dynamic urban wage premium is significant: doubling the population of a commuting zone is associated with a €0.09 increase in hourly wage growth. We then systematically decompose this premium into several components. First, we show that when we control for firm fixed effects, the dynamic urban wage premium falls by 48.5%, suggesting that sorting of firms into cities accounts for a large share of the dynamic urban wage premium. Second, we show that adding detailed controls for coworkers, mirroring the functional forms in Jarosch, Oberfield, and Rossi-Hansberg (2021) and Adenbaum, Babalievsky, and Jungerman (2024), without the firm fixed effects, similarly reduces the dynamic urban wage premium by 46.1%. Adding both these coworker terms and the firm fixed effects accounts for 57.3%.

Next, we provide evidence that what remains of the dynamic premium is concentrated among workers who change jobs, consistent with a job-ladder channel rather than human capital growth. We show that job transitions occur more frequently in big cities: doubling city population is associated with a 2.0 percentage point increase in the rate of job transitions, and a 0.7 percentage point increase in transitions between employers. We then redo our decomposition restricting the sample to workers that did not switch jobs, allowing us to control for these differences across cities. For these job stayers, controlling for firm fixed effects and coworkers leaves a dynamic premium 83.7% below the full-sample baseline.

This is our main finding: differences in firms, coworkers, and job mobility across cities account for the large majority of the urban wage growth premium. The residual for workers who stay in their job is economically negligible: we estimate an upper bound on the ambient effect of city size of €0.015 of additional hourly wage growth per doubling of city size, against a raw dynamic premium of €0.09. Insofar as wages grow faster in bigger cities, our results suggest it arises from differences in distributions of firms and coworkers, as well as heterogeneity in job mobility patterns. These results shed new light on the underlying mechanisms behind the measured urban wage growth premium and leave little room for ambient spillovers "*in the air*".

In this paper, we intentionally do not take a stand on why there is sorting in the first place. There are many potential explanations that have been proposed by the literature and are consistent with our findings, including forces that might reasonably be called agglomeration, but for our purposes, the underlying sources of sorting are not the object of interest. Our finding that firm and coworker composition accounts for most of the urban wage growth premium does not mean that learning is unimportant. It means that learning operates through the concrete channel of who you work with, rather than through diffuse proximity to a large population. Cities matter for wage growth because of the firms and workers that locate there, and because job transitions happen faster. But conditional on these factors, city size itself has little additional explanatory power. Our contribution is to show that whatever generates faster wage growth in larger cities operates through channels that are concrete and measurable: the composition of firms, the composition of coworkers, and the rate of job mobility. This decomposition provides useful guidance for structural modeling, since it suggests that causal relationships between city size and wage growth cannot be treated as primitives.

Related Literature. We contribute to the literature on why wages are higher in bigger cities. Since Glaeser and Maré (2001) it has been known that this premium has a static component (wages are higher) and a dynamic component (wages grow faster). Early work explained the static premium through agglomeration (Ciccone and Hall, 1996). However, more recently, a large literature has argued that it mainly reflects sorting of workers and firms. Combes, Duranton, and Gobillon (2008) show that sorting of workers by itself can explain a large share of regional income differences, and Gaubert (2018) provides evidence that a large share of productivity differences across space are not due to place-specific effects but rather due to firm sorting. Recent estimates vary in how much they attribute to sorting versus location effects (Card, Rothstein, and Yi, 2025), and Butts, Jaworski, and Kitchens (2023) find a growing role for worker sorting in the United States. Carry, Kleinman, and Nimier-David (2026) show that once accounting for sorting of establishments, the elasticity of wages with respect to density falls substantially. These papers argue that the static premium is largely accounted for by the composition of the inputs to production. In our paper, we ask whether the same is true of the dynamic premium.

The literature on the dynamic urban wage growth premium has largely interpreted

it as faster learning in cities. Baum-Snow and Pavan (2012) provide evidence that faster wage growth in big cities is driven by greater returns to experience on the job. De la Roca and Puga (2017) show that workers in bigger cities do not have higher initial unobserved ability. Instead, they accumulate more valuable experience that persists over time and translates to higher wages even after relocating away from the larger cities, despite the sorting interpretation of D'Costa and Overman (2014). Eckert, Hejlesen, and Walsh (2022) use the random allocation of refugees in Denmark to offer causal evidence that the returns to work are higher in bigger cities. Hong (2024) finds that most of the Canadian growth premium accrues to workers who change firms, and that the within-firm remainder is accounted for by firm characteristics. However, these papers do not ask what determines the faster accumulation of human capital in cities. In this paper, we argue that it is necessary to consider the underlying production function for human capital. If workers learn from their firms and their coworkers, then faster rates of skill accumulation in bigger cities may reflect sorting of these underlying inputs to human capital production, rather than diffuse and ambient agglomeration effects. Our contribution to this literature is to provide new and striking estimates that suggest that this is indeed the case, with 57% of the raw dynamic premium accounted for by the composition of firms and coworkers. Our estimates complement the existing literature on the role of job mobility in explaining the dynamic wage premium. Among workers who stay in their job, the premium falls 84% below the full-sample baseline once firms and coworkers are accounted for. These results imply that the existing literature's treatment of faster skill accumulation in big cities as a competing explanation to sorting may require reinterpretation. Indeed, our results suggest that when viewed through the relevant lens, sorting on the inputs to human capital production is a large part of why wages grow faster in big cities.

Our focus on the sorting of the inputs to human capital production is motivated by the literature on the underlying mechanisms through which workers accumulate skills on the job. A growing literature has studied human capital spillovers on the job and found evidence that workers learn from their more-skilled coworkers. Jarosch, Oberfield, and Rossi-Hansberg (2021), Herkenhoff, Lise, Menzio, and Phillips (2024), and Adenbaum, Babalievsky, and Jungerman (2024) all study the sources of human capital growth using structural models and find a major role for learning from higher-skilled coworkers, *within the firm*. Nix (2020) finds that having more educated coworkers

causally increases wage growth. Gregory (2023) finds a role for the firms themselves in driving human capital accumulation. We bring these mechanisms to bear on the dynamic urban wage premium. In particular, we build upon and extend the empirical framework of De la Roca and Puga (2017) in a setting where we directly observe the firm at which a worker is employed and the universe of their coworkers. Our preferred specification nests their dynamic wage equation in first differences, while allowing us to incorporate the firm and coworker effects that the literature on worker skill accumulation suggests are relevant. After controlling for each worker's position relative to their coworkers within the firm, and firm fixed effects, we find that the remaining dynamic premium is concentrated among workers who change jobs.

Our results are of particular relevance to papers that attempt to structurally model the dynamic urban wage premium. Structural models that attempt to explain this premium largely treat it as an agglomeration effect, in which a worker's human capital grows with the overall level of human capital in their city (Duranton and Puga, 2004). Davis and Dingel (2019) introduce a multi-city model where higher productivity in bigger cities is a result of costly idea exchange, and Martellini (2022), Crews (2023) and Lhuillier (2024) consider models of human capital growth in cities where spillovers depend on the skill distribution of the entire population. An exception is Lindenlaub, Oh, and Peters (2024), who show that a steeper firm ladder in more productive cities can provide faster wage growth even without any human capital accumulation. The policy implications of a model where learning spillovers depend on the firms and workers that sort into a city may be quite different from those of a model where those spillovers depend on the size of the city directly. Our findings also provide a set of empirical targets that can be used to discipline models focused on optimal spatial policy (Fajgelbaum and Gaubert, 2020, Bilal and Rossi-Hansberg, 2021, Crews, 2023).

1 Data

We use administrative data from France made available to researchers by the French National Statistical Institute, *Institut national de la statistique et des études économiques* (INSEE). These data are widely used in labor and urban economics and allow us to match workers to their employers over time. Specifically, they are compiled from the *Déclarations annuelles de données sociales* (DADS), mandatory tax forms firms

in France must submit each year detailing information on their employees (such as wages, hours worked, occupation, etc.). Two main datasets are available. First, for the near-universe of workers in France, we have access to a *short panel* that contains information on all firm-worker-year tuples but scrambles worker identifiers every two years. Second, we use the *long panel* which consists of, for a 1-in-12 sample of workers born in October, complete employment histories starting from the worker's first employment spell.

Sample construction. We apply the same sample restrictions to both the short and the long panels, which we balance. We restrict the sample to private-sector employees aged 18–65 working in metropolitan France.[1] When we use the short panel, we use the 2014–2015 cross-sections; the raw long panel covers 1997 through 2019, and our estimation window is 2003–2019, the years over which occupation codes are reliably harmonized. We drop workers with missing information on key variables such as wage, gender, and hours, as well as those with non-valid identifiers which we require for longitudinal analysis. We also drop workers from the sample earning less than €100 per year, and, in both panels, observations with hourly wages outside the interval from 2 to 1,000 euros in either year, which we treat as recording errors (in the cross-sections these bounds remove roughly half a percent of observations). In the long panel, where we track the same worker across years and one-year recording errors would otherwise enter wage growth directly, we additionally drop employed worker-years with annual hours outside 120 to 3,600 and one-year wage spikes that fully reverse the following year (0.18% of worker-years).

Investigating the role of coworker effects in explaining the urban wage growth premium means we also have to (1) define the boundaries of a city and (2) define the boundaries of a firm or a team, i.e. the set of coworkers.

Commuting zones. Our notion of a city is a commuting zone (CZ), which is a set of municipalities that are economically connected. We use the 2010 version of the CZs, which are defined by INSEE and are based on the 1999 census. The CZs are constructed such that they capture the commuting patterns of workers, with the goal of identifying local labor markets. In Figure 1, we plot the size distribution of

[1] This excludes workers employed in overseas territories like Martinique and Guadeloupe.

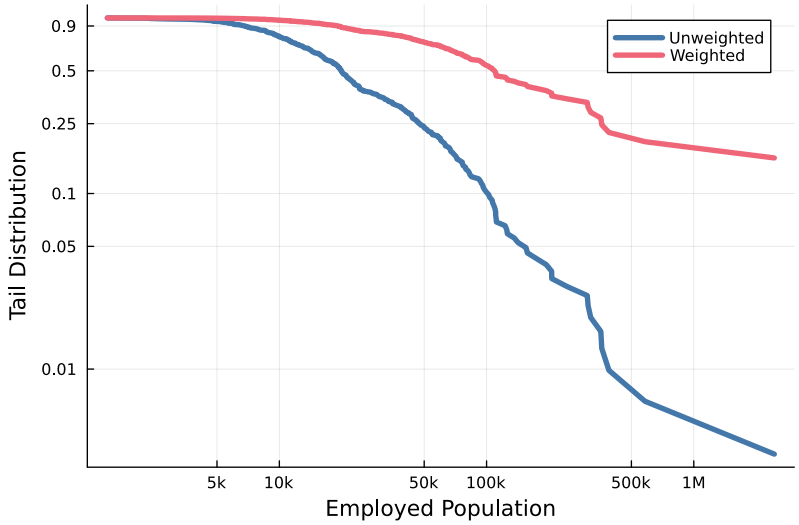


Figure 1: Pareto Tails of the Commuting Zone Size Distribution

Note: This figure plots the complementary cumulative distribution function (tail distribution) of commuting zone size, defined as the share of commuting zones with employed population greater than x, for both the unweighted and employment-weighted commuting zone size distributions. Computed in the 2015 DADS-Postes.

CZs in France, measured by employed population. As expected, the long right tail is capturing the dominant role of Paris as the largest labor market in France.

Defining a team. Following the approach in Adenbaum, Babalievsky, and Jungerman (2024), we identify a team as the intersection of an establishment and a 1-digit occupation code. This definition is intended to capture the set of coworkers that a worker is likely to encounter in their day-to-day work environment.

The choice of team boundaries involves an inherent tradeoff. If defined too narrowly, it will exclude interactions between workers who actually collaborate and share knowledge. If defined too broadly, it will incorporate workers who never meaningfully interact. Our definition takes a conservative stance: we treat all workers at a given establishment within the same 1-digit occupation as potential sources of interactions. We do not further partition teams by additional characteristics like industry classification, sector, or echelon (e.g. entry-level versus manager), so that we can take as broad a stance as possible about the potential learning interactions between workers who may be interacting within an establishment.

Table A.1 presents the 1-digit occupation categories we use, along with selected examples of the more detailed 2-digit and 4-digit codes that fall under *Executives and High-Level Professionals*. If we chose to use finer measures of occupations for our classification of teams, this would imply that workers in closely related roles do not interact. For instance, using either 2 or 4-digit occupation codes would require us to implicitly assume that dentists do not interact in meaningful ways with dental residents, or that lawyers do not interact with other legal professionals. Such granular categorizations would artificially exclude what are likely productive learning relationships within the workplace.

Table 1 presents the transition rates across different organizational groupings in our data. Over the 2014–2015 period, we observe that 73.59% of workers continue with the same team as we have defined it. When we later restrict attention to job stayers, we additionally require full-year employment in the outcome year, which excludes mid-year hires; 80.7% of team stayers meet this requirement, so the stayer estimation sample covers 59.4% of workers on the same population. Under our definition, there are three possible ways a worker can be classified as changing teams: relocating establishments, switching 1-digit occupations, or doing both at the same time. Transitions across establishments account for the bulk of the mobility we document, with approximately 21% of workers moving between establishments year-over-year (not necessarily within the same firm).

The firm-level self flow rate is much higher than that for establishments, much as we would expect (the firm, after all, is a superset of the establishment). The gap reflects workers who remain with the same employer but switch establishments *within the same firm*. Because these sorts of transitions fundamentally change the composition of colleagues with whom a worker interacts on a day-to-day basis, we classify these individuals as having changed teams even though they remain at the same employer.[2]

In Table 2, we provide summary statistics for our team definition, both unweighted and employment-weighted, as well as a comparison to alternative definitions. The unweighted average team size in our sample is 4.75 with a 5% of teams being larger than 14 workers. However, the employment weighted mean is 153 with a long right

[2] We abstract away from the question of whether or not workers are able to interact with their colleagues remotely, since our data window concludes before the onset of the COVID-19 pandemic, and the corresponding widespread adoption of remote working arrangements.

Table 1: Self-Flow Rates

	Rate (%)
1-Digit Occupation	89.55
Firm	83.46
Establishment	78.86
Establishment × 1-Digit Occupation	73.59

Note: This table displays the share of workers who remain in the same organizational unit between 2014 and 2015, computed using the DADS-Postes data.

Table 2: Team Definition Summary Statistics

	Mean	**SD**	**p25**	**Median**	**p75**	**p95**
Unweighted						
Firm Size	8.75	225.45	1.00	1.00	4.00	20.00
Establishment Size	7.53	46.12	1.00	2.00	5.00	25.00
Team Size (Establishment × 1-Digit Occupation)	4.75	26.58	1.00	1.00	3.00	14.00
Employment-weighted						
Firm Size	5,819.53	19,995.25	13.00	105.00	1,064.00	44,951.00
Establishment Size	289.90	1,156.48	9.00	39.00	173.00	1,016.00
Team Size (Establishment × 1-Digit Occupation)	153.33	635.08	4.00	18.00	89.00	549.00
1-Digit Occupations per Establishment	3.15	1.26	2.00	3.00	4.00	5.00

Note: Summary statistics of team size under our preferred definition (establishment interacted with 1-digit occupation) and alternative organizational boundaries, presented both with and without employment weights. Source: 2015 DADS-Postes.

tail. For instance, the 75th percentile is 89 and the 95th percentile is 549. Finally, we compute the mean number of 1-digit occupations per establishment, which is 3, suggesting that establishments on average have a very diverse set of workers.

2 Decomposing the Urban Wage Premium

In this section, we unpack the dynamic urban wage premium in three steps. First, and most crucially, we show that this dynamic premium attenuates substantially when we control for firm fixed effects and coworker controls. Next, we provide evidence that job mobility is higher in bigger cities in France, and finally, we show that the attenuation of the urban wage premium is even more pronounced when we restrict attention to workers who stay in the same job. Taken together, these results provide a concrete statistical explanation for the urban wage premium: it is largely accounted for by the

sorting of workers to firms and coworkers, and what remains is concentrated among movers and accompanied by greater mobility in larger cities. This leaves little room for canonical ambient knowledge spillovers.

2.1 Main Results

We now turn to the main analysis. In a first stage, we regress wages on their own lag and commuting zone fixed effects for workers who are employed in both periods of our short panel and stay in the same commuting zone. We estimate

$$w_{i,t} = \nu w_{i,t-1} + \psi_{c(i,t)} + \epsilon_{i,t} \tag{1}$$

where $w_{i,t}$ is the hourly wage of worker i in commuting zone $c(i,t)$ at time t[3], and $\psi_{c(i,t)}$ is a commuting zone fixed effect. The coefficient ν captures the persistence of wages[4], and the fixed effect $\psi_{c(i,t)}$ captures the commuting-zone component of wages conditional on the lagged wage, that is, differences in wage growth across zones among workers with the same lagged wage. Moreover, we show in Appendix C that this specification formally nests the dynamic specification of De la Roca and Puga (2017) when we restrict attention to workers who have not changed jobs, which will be our preferred specification, as discussed in Section 2.3.[5] Our main specifications will use hourly wages in levels, however, all of our results are robust to using logged hourly wages as the dependent variable.[6] We report these results in Appendix D.1.

[3] Throughout the paper, we use commuting zone to mean the commuting zone of work. By construction of the commuting zones, residence commuting zone differs from work commuting zone for only a very small fraction of workers.

[4] Note that this specification allows us to test whether past wages matter for future wage growth. Subtracting by $w_{i,t-1}$ on both sides and rearranging yields

$$w_{i,t} - w_{i,t-1} = \psi_{c(i,t)} + (\nu - 1)w_{i,t-1} + \epsilon_{i,t}$$

If $\nu = 1$, then wages follow a random walk. If $\nu < 1$, then there is mean reversion. Finally, if $\nu > 1$ then the initial advantages associated with being a high-wage worker compound over time.

[5] We focus on measuring the urban wage *growth* premium, as opposed to the premium in levels. Several papers (Combes, Duranton, and Gobillon, 2008, Carry, Kleinman, and Nimier-David, 2026) have found that the differences in levels are well accounted for by firm and worker sorting, and the results in Glaeser and Maré (2001) and Eckert, Hejlesen, and Walsh (2022) suggest that the static premium is not as robust as the dynamic premium.

[6] It is not obvious *a priori* whether the more natural specification is in levels or logs. Historically, the empirical literature has preferred logs, however, as noted in Borovičková and Shimer (2025), most models of labor search admit a wage equation that is linear in the levels. To stay consistent with this theoretical literature, we thus present our main results in levels.

In the second stage, we collapse the estimated fixed effects from Equation (1) to commuting-zone means $\hat{\psi}_c$ and regress them on the log population p_c of commuting zone c, one observation per zone:

$$\hat{\psi}_c = \alpha_0 + \alpha p_c + \eta_c \tag{2}$$

We estimate Equation (2) by OLS across the 304 commuting zones with heteroskedasticity-robust standard errors.

By projecting the estimated fixed effects onto population in the second stage, we can quantify how much of the cross-commuting zone wage growth variation is systematically related to commuting zone size versus idiosyncratic commuting zone-specific factors. The coefficient α in the second stage tells us the semi-elasticity of hourly wages with respect to city population, while the R^2 reveals what fraction of the cross-city wage differences can be explained by population alone. As we progressively add controls in subsequent specifications, changes in both α and R^2 will reveal which mechanisms drive the urban wage premium.

The results for our baseline specification are reported in Column (1) of Table 3. This specification captures the raw urban wage growth premium that we see in the data, without any further controls, and we will refer back to it for the rest of the analysis as we seek to understand which channels and mechanisms drive the effect. In our second stage, we find a semi-elasticity of wages with respect to city size of 0.14. Taken at face value, this would suggest that doubling the population in a commuting zone is correlated with a €0.09 increase in hourly wage growth, year over year. This is a relatively large effect, of the same order as the results found in De la Roca and Puga (2017) in Spain over a similar time period.[7] The R^2 of our second stage regression is relatively high: log population alone explains 29.7% of the raw differences in average wage growth across commuting zones.

In what follows, we add firm effects, coworker controls, together and separately, to the baseline specification to examine how much of the measured urban wage growth

[7] De la Roca and Puga (2017) estimate a short-term elasticity of earnings with respect to city size of 0.0223 and a medium-term elasticity, after 7.7 years of local experience, of 0.0510 (their Table 2). The difference implies that a year of experience in a city twice as large raises earnings by about 0.0037 log points per log point of population. Our baseline estimated in log hourly wages, which is directly comparable, is of the same order (Table D.1, Column (1)).

Table 3: Decomposing the Urban Wage Premium

	(1)	(2)	(3)	(4)
	Baseline	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions				
Lagged Wage (ν)	0.9140	0.8115	0.9597	0.7944
	(0.0203)	(0.0393)	(0.0157)	(0.0530)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)			0.2146	0.0948
			(0.0067)	(0.0521)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)			0.0154	-0.1521
			(0.0370)	(0.0439)
Nonlinear Effects ($\tilde{\theta}_2^+$)			-0.0004	0.0003
			(0.0000)	(0.0004)
Nonlinear Effects ($\tilde{\theta}_2^-$)			-0.0005	-0.0004
			(0.0001)	(0.0000)
Lagged Firm Fixed Effects	No	Yes	No	Yes
Observations	16,628,524	14,953,462	14,698,578	14,698,578
R^2	0.7177	0.7528	0.7148	0.7560
Panel B: Second Stage Regressions				
Log Population (α)	0.13579	0.06999	0.07326	0.05799
	(0.0164)	(0.0158)	(0.0112)	(0.0140)
Observations	304	304	304	304
R^2	0.2965	0.1113	0.1302	0.0840
$\hat{\alpha}$ change relative to (1)	–	-48.5%	-46.1%	-57.3%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wages. Column (1) shows the baseline specification from Equation (1) with commuting zone fixed effects and lagged wages. Column (2) adds firm fixed effects following Equation (3). Column (3) includes the full coworker effects specification from Equation (4). Column (4) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust. Sample consists of all workers in the short panel with residualized wages for age and gender.

premium is statistically accounted for by firm and coworker composition. A natural concern is that workers sort across cities on unobserved growth potential, so that part of what we attribute to firms and coworkers instead reflects characteristics of the workers themselves. We cannot include worker fixed effects in the short panel, because worker identifiers are scrambled every two years, so each worker contributes a single one-year transition. The formal econometric justification for how we address this concern is in Appendix B, but intuitively the key assumption is that workers in bigger cities are not negatively selected on earnings growth potential conditional on covariates. This assumption is natural and is justified by an extensive literature showing that workers in bigger cities are positively selected on skill (Combes, Duranton, and Gobillon, 2008, Carry, Kleinman, and Nimier-David, 2026, Rouwendal and Koster, 2025). Under this assumption, the city-size gradients we report are upper bounds on the ambient effect. To the degree that worker sorting biases them, it raises the measured gradient rather than lowering it, which would only strengthen the claim that the ambient effect is small. In the long panel, where we follow workers for up to seventeen years, we also add worker fixed effects and find the same qualitative pattern (Appendix D.6).

For our first set of expanded controls, we add lagged firm fixed effects and estimate the equation

$$w_{i,t} = \psi_{c(i,t)} + \mu_{j(i,t-1)} + \nu w_{i,t-1} + \epsilon_{i,t} \tag{3}$$

where we define the firm $j(i, t-1)$ to be everyone with the same 1-digit occupation code and the same employer.[8] The firm fixed effects capture any invariant characteristics of the firms which may be relevant for wage growth at the worker level, including the industry, the type of work they are engaged in, and other invariant characteristics of their workforce composition. For instance, if a firm in our sample consistently has a better composition of coworkers, then this will be absorbed in the firm effect.[9]

[8] Notice that our definition of a firm here differs from our definition of a team or set of coworkers described in Section 1. We need to use firm IDs and not establishment IDs here because we are already including commuting zone fixed effects. We cannot separately identify establishment times occupation codes fixed effects from commuting zone fixed effects.

[9] Unlike in Abowd, Kramarz, and Margolis (1999), who aim to jointly identify firm and worker fixed effects, we do not see the full sample of workers for a long enough panel to allow for credible identification of the worker fixed effects. Instead, by including the lagged wage in our specification, we nest the case where we estimate the model in first differences, and therefore are only concerned with identification of the *firm* effects. These are pinned down by the average growth rate of wages within the firm. Because the firm effects enter alongside the commuting zone effects, the two are

As before, in the second stage we project down the estimated commuting zone fixed effects onto log population. The results are reported in Column (2) of Table 3.

Having added firm fixed effects, it is striking to examine our measure of the urban wage growth premium (α). We see that the semi-elasticity of hourly wages with respect to commuting zone population falls from 0.14 to 0.07, a drop of 48.5%. By adding firm fixed effects, we account for the fact that the composition of firms varies between big cities and small cities. In Table 4 we regress the employment-weighted average firm fixed effect by commuting zone, from the specification with both firm effects and coworker controls (Column (4) of Table 3), on log population. We see that high wage growth firms tend to be located in larger cities. This underlying pattern of firm sorting confounds the raw estimates of the urban wage growth premium, and causes it to overstate the importance of population size per se on the growth rate of wages. We see as well that after controlling for firm effects, population becomes dramatically less informative in explaining average changes in wages across commuting zones, only accounting for 11.1% of the variation (compared to 29.7% in our baseline case).

Table 4: Projecting Firm Fixed Effects on Commuting Zone Size

	(1)
Dependent variable	Firm Fixed Effect
Log Population	0.1435
	(0.0132)
Observations	304
R^2	0.2942

Note: This table reports regressions of firm fixed effects on commuting zone size. The dependent variable is the employment-weighted commuting-zone mean of the firm fixed effects estimated in Column (4) of Table 3. Robust standard errors in parentheses.

separately identified through firms that operate in more than one commuting zone (Lemma 2 in Appendix B). The specifications also differ in their estimation samples, and we show below that re-estimating the baseline on the common sample leaves the attenuation essentially unchanged. The coworker specification of Equation (4) does not rely on such firms, and attenuates the gradient by a similar amount.

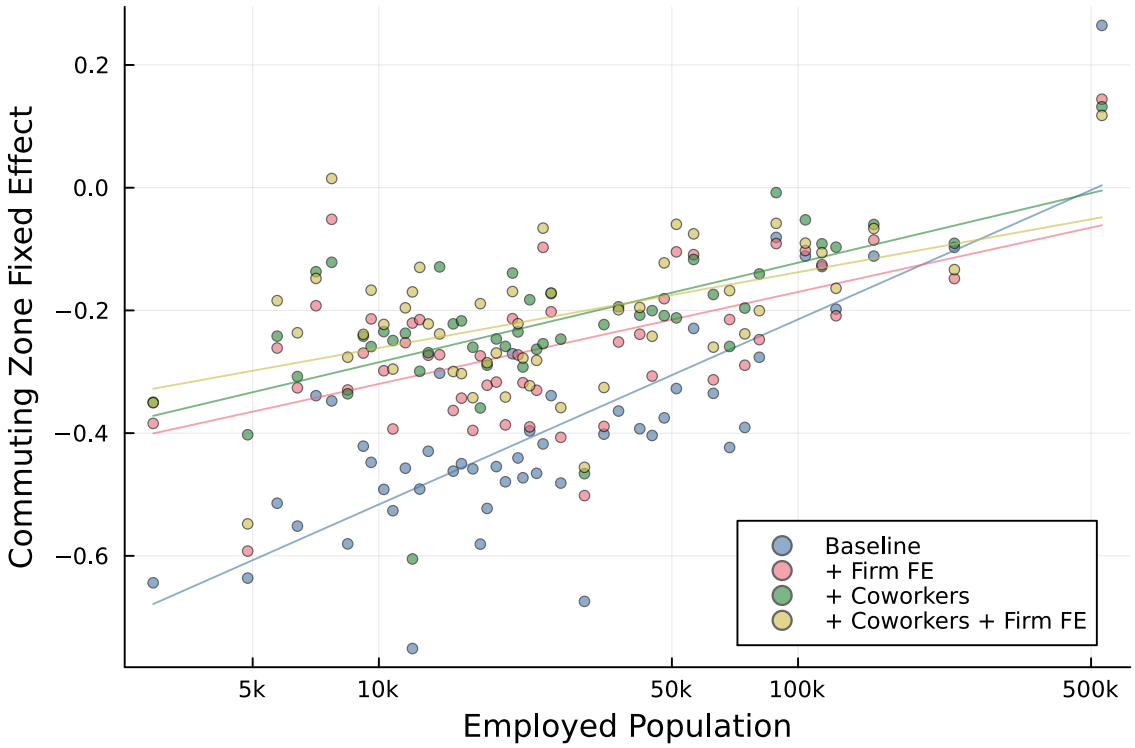


Figure 2: City Fixed Effects vs Population

Note: This figure plots the average commuting zone fixed effects for each 2% quantile bin of commuting zone log population. "Baseline" plots the CZ fixed effects from Equation (1). "+ Firm FE" plots the CZ fixed effects from Equation (3). "+ Coworkers" plots the CZ fixed effects from Equation (4). "+ Coworkers + Firm FE" plots the CZ fixed effects from Equation (5).

Firm fixed effects capture time-invariant firm characteristics, but recent evidence suggests that much of the learning that takes place within firms comes from coworkers. Adenbaum, Babalievsky, and Jungerman (2024) decompose the sources of human capital growth on the job, and find that peer effects account for more than 50% of the variation in human capital growth for workers in France. Of particular importance is the effect of more skilled coworkers, which contributes 52.5% of the variation in human capital growth, even while accounting for workplace and individual effects. Moreover, unlike firm fixed effects, which capture invariant features that are common across all their establishments, the composition of one's coworkers is quintessentially local. If the composition of coworkers varies substantially across commuting zones, due perhaps to the endogenous mobility choices of workers, small differences in the growth rates of skills across cities, or firm sorting behavior, then we should expect that the opportunities to learn from coworkers will vary with city size.

To test this hypothesis, we augment our baseline specification with controls for peer effects, as in Adenbaum, Babalievsky, and Jungerman (2024) and Jarosch, Oberfield, and Rossi-Hansberg (2021). For each worker i in our sample, we identify the set $\mathbb{W}^+_{i,t}$ of i's coworkers who are paid more than i, and the set $\mathbb{W}^-_{i,t}$ of coworkers who are paid less than i. We compute over each of these sets the average deviation in wages from worker i's wage, and the average squared deviation (which captures non-linearity in the learning function).[10] Consistent with our approach here, we identify the relevant set of coworkers in the previous period $t-1$, since the wage growth we see between $t-1$ and t should reflect human capital accumulation in the previous period, with the coworkers at time $t-1$ rather than the present-day coworkers. Our choice to separately control for the effects of higher and lower paid workers reflects the findings in Herkenhoff, Lise, Menzio, and Phillips (2024) that the human capital production function is nonlinear, with peer effects that depend on a coworker's relative position

[10] It is not obvious a priori how to compute these coworker controls in a computationally feasible way, since the naive algorithm's cost is quadratic in the size of the dataset. In Appendix A.2 we show how to compute these coworker controls in linear time by carefully rewriting the problem using running sums.

to the worker in question. We then estimate the equation

$$w_{i,t} = \psi_{c(i,t)} + \nu w_{i,t-1} + \underbrace{\tilde{\theta}_1^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Higher-Wage Coworkers}} + \underbrace{\tilde{\theta}_1^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Lower-Wage Coworkers}}$$
$$\underbrace{+ \tilde{\theta}_2^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1} + \tilde{\theta}_2^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1}}_{\text{Nonlinear Effects}} + \epsilon_{i,t} \tag{4}$$

Note that in this regression equation, we do not yet include firm fixed effects. We see that our measure of the urban wage growth premium declines by a comparable amount in this specification, with the semi-elasticity of hourly wages with respect to commuting zone population falling from 0.14 in our baseline case to 0.07 with coworker effects, a drop of 46.1%. This is of a similar magnitude to the decline that we see in Equation (3), which we interpret as an indication that our coworker controls are capturing much of the same variation as the firm fixed effects. The quality of a firm's workforce (and the corresponding potential for learning from one's coworkers) is common across all the workers at the firm. However, our coworker controls allow us to capture the fact that individual workers within the firm may have different potential for future human capital growth depending on their position within the within-firm human capital distribution. For example, the most skilled worker in a firm has no one else to learn from, and our coworker controls can capture this.

In principle, we would like to account for both these sorts of coworker controls *and* for invariant firm effects. To do so, we repeat the coworker specification from Equation (4) adding in lagged firm FE.

$$w_{i,t} = \psi_{c(i,t)} + \nu w_{i,t-1} + \mu_{j(i,t-1)} + \underbrace{\tilde{\theta}_1^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Higher-Wage Coworkers}} + \underbrace{\tilde{\theta}_1^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Lower-Wage Coworkers}}$$
$$\underbrace{+ \tilde{\theta}_2^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1} + \tilde{\theta}_2^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1}}_{\text{Nonlinear Effects}} + \epsilon_{i,t} \tag{5}$$

It is worth taking a moment to discuss the question of separate identification of the effects here. In principle, we can separately identify coworker and firm effects off of the average growth rate of wages by firm, and the gradient of wage growth *within the firm* against a worker's position within the wage distribution of the firm. That is, if workers who are lower in the wage distribution consistently grow faster than the firm average, then this will be attributed to $\tilde{\theta}_1^+$. In practice, we expect that these effects are likely to be highly collinear with the firm effects, and difficult to disentangle statistically. However, for our purposes this does not pose a substantial problem: we are not interested in assigning any significant interpretation to the coefficients of our coworker controls. Rather, our object of interest is $\hat{\alpha}$: the correlation between the commuting zone fixed effects $\psi_{c(i,t)}$ and the population of the commuting zone $p_{c(i,t)}$, after partialling out the effects of our other control variables.

This specification also highlights why we do not estimate the full dynamic and static wage specification from De la Roca and Puga (2017) directly, and instead use an estimating equation that coincides with theirs in first differences. Their approach relies on the use of city size bins in order to make the estimation procedure tractable. There is no natural generalization of this to firm fixed effects, or to the coworker learning effects that we are interested in. In particular, it is difficult to imagine how one would capture the *history* of coworker effects in a way that maps neatly onto their experience terms. For this reason, we restrict attention to our levels specification with the lagged wage as a free regressor, whose $\nu = 1$ restriction coincides with their specification in first differences (Appendix D.3), and where the generalization to firm and coworker controls is clear.

We can see the results from this specification in Column (4) of Table 3. We find that with *both* coworker controls and lagged firm fixed effects, the measured urban wage growth premium is somewhat smaller than with coworker effects alone: the semi-elasticity drops to 0.058, a 57.3% fall from our baseline. Another way to report these results is to plot the average commuting zone fixed effects for various quantile bins of log population, for each specification. We do this in Figure 2, and we see that the slope of the line of best fit decreases dramatically when we add either firm fixed effects or coworker controls. The commuting zone fixed effects look fairly similar controlling for either firm fixed effects or coworker controls, suggesting that these two controls are capturing very similar variation in the underlying quality of the

learning environments across firms. It is telling that when we compare the fixed effects controlling for coworkers (green) versus coworkers and firm fixed effects (gold), the results are almost indistinguishable to the eye. This reflects the fact that our measures of the urban wage growth premium (α) are statistically indistinguishable between Columns (3) and (4) in Table 3.

The estimation samples shrink across the columns of Table 3: workers in single-person lagged teams have no defined coworker controls, and workers with a missing lagged firm identifier drop from the firm specifications, so Column (1) uses 16.6 million observations while Column (4) uses 14.7 million. To verify that the attenuation does not reflect composition across estimation samples, we re-estimate the baseline on exactly the Column (4) sample. The baseline gradient there is 0.147, slightly above the full-sample 0.136, so the attenuation computed within a single population is 60.5% rather than 57.3%. The comparison across columns, if anything, understates the attenuation on a common sample.

As shown in Appendix B.3, adding controls need not sequentially attenuate our coefficient of interest. Changing the controls in the first stage changes both the channels absorbed by the regression and the sorting that remains. The observed decline in the measured urban wage growth premium indicates that the firm and coworker controls absorb wage-growth components positively associated with city size. For example, starting from the baseline, adding coworker controls could have increased the coefficient if coworker controls were correlated with a dimension of wage growth that is negatively associated with city size. The fact that the coefficient declines indicates that this is not the case, and this holds for each addition of controls. Moreover, adding these controls can change other coefficients, but this is innocuous for our purposes as we are only interested in the coefficient on commuting zone fixed effects (see Appendix B.4 for discussion).

Robustness. We relegate several robustness checks to the appendix. First, in Appendix D.1 we repeat the analysis for logged hourly wages. The results are reported in Table D.1. These results show extremely similar qualitative patterns to what we find in our main specification. In fact, in log wages, the results are more striking, with a 90% decline relative to our baseline specification when we restrict to job stayers and add coworker controls and firm fixed effects. Second, in Appendix D.3, we repeat the

analysis in first differences, which restricts the persistence coefficient to $\nu = 1$. For example, Equation (1) becomes

$$w_{i,t} - w_{i,t-1} = \psi_{c(i,t)} + \epsilon_{i,t} \tag{1'}$$

The results are reported in Table D.6. The firm channel carries through: the first-differenced premium falls by 33.4% with firm fixed effects and by 33.3% with firm fixed effects and coworker controls together. The coworker controls alone reduce the first-differenced premium by only 14.6%, which is expected: with ν restricted to one, transitory wage reversion loads onto the coworker terms rather than the lagged wage, which is one of the reasons the freely estimated AR(1) specification is our main specification. Among job stayers the fully controlled first-differenced premium again vanishes (a 99.9% decline; the projected slope is statistically indistinguishable from zero).

Third, in Appendix D.2 we winsorize the euro wage tails and find the results essentially unchanged: the baseline gradient moves from 0.136 to 0.106 and the fully controlled gradient from 0.058 to 0.053, so the euro results are not driven by the remaining tails. Fourth, in Appendix D.5 we repeat the main decomposition in the 2013, 2014, 2016, and 2017 cross-sections and, if anything, the attenuation is larger in these other years. Finally, in Appendix D.4, we repeat the analysis using alternative measures of the within firm wage distribution to explore how much of our coworker controls can be explained by standard moments. In particular, we consider the mean, variance, skewness, kurtosis, and other percentiles of the within firm wage distribution (the 1st, 10th, 90th, and 99th percentiles). Consistent with our previous specifications, where we allow the effect for a given worker to depend on their relative position within the wage distribution, we introduce all of these controls for the distribution of coworkers and allow for interaction terms with the worker's wage. The results are reported in Table D.8. With the full battery of moment controls, our measure of the urban wage growth premium α falls by 58.5% relative to our baseline, compared to 46.1% with our preferred coworker controls. Hence, our results are not driven by our choice of functional form for coworker interactions.

Interpretation. These results suggest that insofar as we observe an urban wage growth premium among workers in France, around 57% is accounted for by differences

in the composition of both firms and coworkers across urban areas. This underlying mechanism in principle is a combination of several things: First, workers could experience wage growth because their skills have increased through learning. Second, workers' wages could increase because they have switched jobs, and this has allowed them to move up the wage ladder. Third, workers' wages could have increased in their current job for various non-learning reasons (e.g., if an outside offer gives them leverage to bid up their current wages).

Our learning controls and firm effects speak directly to the first mechanism. However, this approach to measuring the urban wage growth premium conflates the effects of learning with the effects of job to job transitions. If workers transition jobs faster in larger cities, then mechanically we would observe this as a correlation between the commuting zone fixed effects and population. In the next section, we investigate whether or not this is the case: that is, do workers in bigger cities change jobs faster?

2.2 Employment Transitions

To investigate the question of whether or not the job transition rates differ across cities, in Equation (6), we run a simple regression of job transition rates as a function of log city population:

$$EE_{i,t} = \alpha_0 + \alpha p_{c(i,t)} + \epsilon_{i,t} \tag{6}$$

We use a cross-section from the short panel, which records a worker's employer in the current period and in the lagged period. Note that in this case, our notion of the employer is the same as we use for our definition of the set of coworkers: we identify a worker as experiencing a job transition if they change *establishments or 1-digit occupations* from one year to the next. This means that we treat both internal movements between establishments within a firm, or changes in roles large enough to change the worker's 1-digit occupation code, as *job transitions*. For our purposes, such changes in the worker's team within the firm are very similar to external moves, in terms of the expected effect they will have on wages. This broad measure is not employer-to-employer mobility in the sense of the search literature, because it includes within-firm across-occupation mobility. Therefore, we also construct a more standard measure of job transitions, which is a change of employer between the two years. We report the results in Column (1) of Table 5 for the broad measure and in Column (3) for the change of employer, and find a significant and positive relationship between

city size and job transition rates for both.

Next, in Equation 7 we add a control for lagged wage, i.e., the wage in the first period of the two-period short panel:

$$EE_{i,t} = \alpha_0 + \alpha p_{c(i,t)} + \beta w_{i,t-1} + \epsilon_{i,t} \tag{7}$$

This regression provides evidence on whether workers leave a job of a given quality more or less quickly. This may be driven by better matching in thicker labor markets, or by the higher prevalence of better jobs in bigger cities that can win poaching fights more easily. We report the results in Columns (2) and (4) of Table 5 and again find a significant and positive city size effect. As before, in Table D.3 we repeat the analysis for logged hourly wages and find similar results.

Table 5: Job Transition Rates and City Size

	(1)	(2)	(3)	(4)
Change of	Team	Team	Employer	Employer
Log Population	0.0294	0.0395	0.0105	0.0230
	(0.0018)	(0.0016)	(0.0018)	(0.0013)
Lagged Hourly Wage		-0.0026		-0.0032
		(0.0004)		(0.0006)
Controls	No	Lagged Wage	No	Lagged Wage
Fixed Effects	No	No	No	No
SE Clustering	CZ	CZ	CZ	CZ
Observations	16,628,524	16,628,524	16,628,524	16,628,524
R^2	0.0022	0.0074	0.0004	0.0118

Note: In Columns (1)–(2) the outcome is a change of team (establishment × 1-digit occupation) between the two years and in Columns (3)–(4) it is a change of employer (firm). Columns (1) and (3) show the baseline specification from Equation (6). Columns (2) and (4) add lagged hourly wages as a control following Equation (7). Standard errors are clustered at the commuting zone (CZ) level. The sample consists of all workers from one cross-section of the short panel.

These results stand in contrast to Martellini (2022) and Hong (2024), who find little evidence for faster job transitions in bigger cities. Understanding the source of this discrepancy is an important goal for future work. It is also important to note that,

while the results support the idea that the unexplained remainder of the dynamic urban wage premium might be explained by differences in the rate of EE moves, this is not necessarily evidence of increasing returns to search. As Lindenlaub, Oh, and Peters (2024) show, the returns to EE transitions may be higher because of positive firm sorting towards more productive cities. Hence, while we provide evidence that labor market transitions can help explain what is left of the dynamic urban wage premium after we control for firms and coworkers, more work is still required to understand exactly what drives this faster climb up the job ladder in bigger cities.

2.3 Job Stayers

Motivated by our evidence from the previous section that larger cities in France appear to have a higher job-to-job transition rate, we now restrict our attention to the subsample of workers who have not transitioned jobs in the past year. This restriction allows us to separate out the effect of faster transitions from faster skill growth because job stayers, by definition, cannot have moved up on the job ladder within the previous year. Changes in their wages must reflect either learning or bargaining.

We re-estimate Equations (1) and (3) to (5) on this subsample and report the results in Table 6. We see a smaller baseline urban wage growth premium among stayers, with a measured semi-elasticity of 0.078 relative to 0.136 in the full sample. Adding firm fixed effects, the growth premium falls to 0.030. Controlling for coworker effects it falls to 0.049, and controlling for both it falls to 0.022. The fully controlled stayer gradient thus sits 83.7% below the full-sample baseline, and 71.6% below the stayer baseline. The remaining coefficient, while statistically distinguishable from zero, is economically small: it implies at most €0.015 of additional hourly wage growth per doubling of city size, compared to our baseline estimate of €0.09. In Figure 3 we plot the commuting zone fixed effects against log population. As before, we check that these results are robust to a variety of other specifications, reported in Appendix D.1. The stayer result also holds, and is if anything stronger, in every other year of the data: across the 2013–2017 cross-sections, the fully controlled stayer gradient sits between 84% and 96% below each year's full-sample baseline, and our headline year is the most conservative specification (Table D.9).

The cross-section results above use two-year panels, so one may worry that they reflect

Table 6: The Urban Wage Premium Holding Job Mobility Constant

	(1)	(2)	(3)	(4)	(5)
	Baseline (Full Sample)	Baseline (Stayers)	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions					
Lagged Wage (ν)	0.9140	0.9780	0.9269	0.9610	0.7800
	(0.0203)	(0.0245)	(0.0402)	(0.0218)	(0.0509)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)				0.1814	0.0727
				(0.0097)	(0.0423)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)				-0.1034	-0.3213
				(0.0305)	(0.0526)
Nonlinear Effects ($\tilde{\theta}_2^+$)				-0.0002	0.0011
				(0.0001)	(0.0006)
Nonlinear Effects ($\tilde{\theta}_2^-$)				-0.0005	-0.0004
				(0.0001)	(0.0000)
Lagged Firm Fixed Effects	No	No	Yes	No	Yes
Observations	16,628,524	9,877,713	8,779,943	8,844,662	8,644,021
R^2	0.7177	0.8384	0.8717	0.8359	0.8757
Panel B: Second Stage Regressions					
Log Population (α)	0.13579	0.07815	0.03007	0.04922	0.02216
	(0.0164)	(0.0130)	(0.0084)	(0.0122)	(0.0078)
Observations	304	304	304	304	304
R^2	0.2965	0.0876	0.0494	0.0285	0.0236
$\hat{\alpha}$ change relative to (1)	–	-42.4%	-77.9%	-63.8%	-83.7%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wages. Column (1) shows the baseline specification from Equation (1) for the full sample. Columns (2)-(5) restrict to job stayers. Column (2) shows the baseline specification from Equation (1) restricting the sample to job stayers. Column (3) adds firm fixed effects following Equation (3). Column (4) includes the full coworker effects specification from Equation (4). Column (5) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust. Wages are residualized for age and gender.

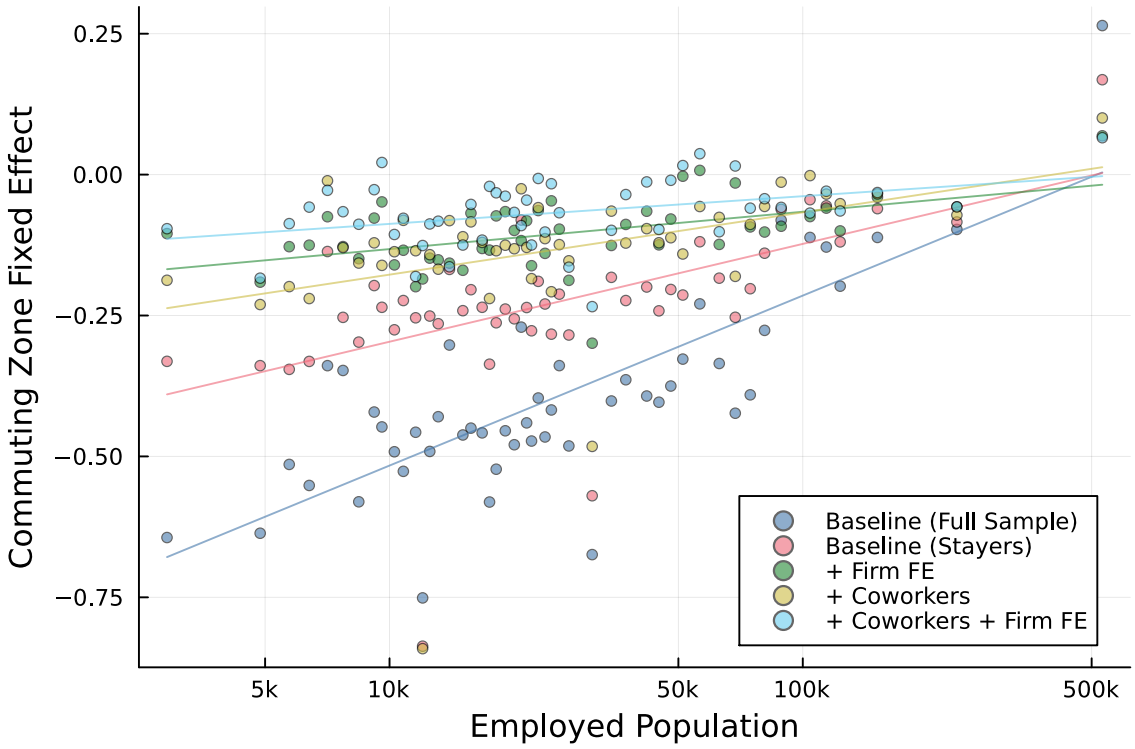


Figure 3: City Fixed Effects vs Population for Job Stayers

Note: This figure plots the average commuting zone fixed effects for each 2% quantile bin of commuting zone log population, in our sample restricted to job stayers. "Baseline (Full Sample)" plots the CZ fixed effects from Equation (1) for the full sample. "Baseline (Stayers)" plots the CZ fixed effects from Equation (1) when we restrict to job stayers. "+ Firm FE" plots the CZ fixed effects from Equation (3) for job stayers. "+ Coworkers" plots the CZ fixed effects from Equation (4) for job stayers. "+ Coworkers + Firm FE" plots the CZ fixed effects from Equation (5) for job stayers.

something particular about single year-pairs. In Appendix D.6 we repeat the entire decomposition on the 1-in-12 long panel, following the same workers over 2003–2019 and adding job-stayer, worker-fixed-effects, and annual-earnings specifications. We also consider several different measures of income: hourly wages, log hourly wages, wage ranks, total earnings, and log total earnings. The lifetime panel reproduces the two central findings: the premium attenuates strongly under firm and coworker controls in all these five wage measures we examine (by 47% to 92%), and the fully controlled premium among job stayers is statistically indistinguishable from zero in four of the five measures, including our main hourly-wage measure. The panel also reproduces the transition-rate gradient of Section 2.2, with a positive and significant job-to-job gradient in every year from 2004 to 2019.

2.4 Discussion

So what exactly is "in the air" that makes wages grow faster in bigger cities? Our results provide a surprisingly concrete answer: when we focus on workers who remain in the same job, thereby eliminating the job ladder mechanisms, the urban wage growth premium falls by 83.7% once we account for firms and coworkers. The residual semi-elasticity is economically negligible, bounding any ambient component of wage growth among stayers at €0.015 per hour per doubling of city size. This suggests that the well-documented urban wage growth premium operates through three specific channels: the quality of firms that locate in cities, the composition of coworkers within those firms, and the speed at which workers can climb the job ladder.

This decomposition helps clarify the mechanisms underlying a phenomenon that has been central to urban economics since Marshall (1890). Rather than diffuse knowledge spillovers that benefit all workers in a city, our findings suggest that faster wage growth is remarkably localized. It depends on the specific firm you work for and the specific people you work with. For job stayers, once we know their firm and coworkers, knowing that they work in Paris versus a small town adds virtually no additional information about their wage growth prospects. This aligns with recent structural evidence on the importance of learning interactions *within the firm*, as in Adenbaum, Babalievsky, and Jungerman (2024), Herkenhoff, Lise, Menzio, and Phillips (2024), and Jarosch, Oberfield, and Rossi-Hansberg (2021). Our results are consistent with the finding of De la Roca and Puga (2017) that the value of big-city experience

persists after workers leave. Human capital acquired from one's firm and coworkers travels with the worker, just as human capital acquired from the city would, so what our results change is the account of where that learning comes from. In all of these frameworks, human capital accumulation operates through direct interactions with specific coworkers rather than through ambient effects.

The three channels we identify (firm quality, coworker composition, and job mobility) together substantially account for the observed urban wage growth premium. Our evidence on sorting shows that higher-quality firms concentrate in larger cities, creating a superior learning environment *within the firm*. Meanwhile, our finding of higher EE job transition rates in larger cities suggests that the job ladder operates more quickly there, whether due to thicker labor markets that facilitate better matches or a steeper firm quality distribution as in Lindenlaub, Oh, and Peters (2024). What's notable is how cleanly these mechanisms separate: for job stayers, the firm and coworker channels account for the large majority of the premium, while the part they do not account for in the full sample is concentrated among workers who change jobs. Researchers who are interested in modeling the dynamics of wage growth across space need not resort to ambient spillovers operating outside these channels, which are difficult to interpret as structural and microfounded mechanisms.

Our findings offer a clearer picture of how urban wage dynamics actually work. By decomposing the urban premium into its constituent parts, we can see that the advantages of cities for wage growth are remarkably concrete and measurable. For workers who remain in the same job, city size itself plays at most a negligible direct role in wage growth once we account for who they work with. While understanding the deeper mechanisms that drive firms and workers to sort into cities, and exactly what generates faster job transitions there, remains an important area for future research, our decomposition provides sharp empirical targets for such work.[11] The mechanisms behind urban wage growth turn out to be more immediate and tangible than the metaphor of knowledge "in the air" might suggest. They operate through the specific people you work with and the specific opportunities available to change jobs. In this fundamental sense, there really is nothing "in the air."

[11] There are several candidate explanations for why firms and workers sort differentially into larger cities, including increasing returns to search in thick markets (Martellini and Menzio 2020), the shape of the job ladder (Lindenlaub, Oh, and Peters 2024), and the role of cities as providers of non-tradable amenities that attract high earners (Lee 2010).

Conclusion

In this paper, we have used French administrative data to provide new evidence on the causes of the dynamic urban wage premium. We first demonstrated that this dynamic premium is substantially attenuated after adding firm fixed effects, suggesting that part of this premium is driven by the firms that choose to sort themselves into bigger cities. Next, we showed that the premium shrinks even further when adding controls for coworker wages, which suggests a role for learning from the higher human capital workers that sort themselves into cities. We also showed that this accounting is robust to how the coworker distribution is parametrized. Combining firm and coworker controls greatly attenuated the dynamic urban wage premium in our full sample, and left only an economically negligible remainder in our subsample of job stayers. Motivated by this contrast between job stayers and the full sample, we showed that job transition rates, including strict employer-to-employer transitions, are higher in bigger French cities. This suggests that labor market factors (whether they are increasing returns to search or a steeper firm quality ladder) account for the remaining part of the dynamic urban wage premium that survived our coworker and firm controls. Taken at face value, our results leave little room for the canonical explanation for the dynamic urban wage premium, namely human capital spillovers "*in the air*".

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A Data Appendix

A.1 Variable List

– Hourly Wages: We use hourly wages to control for variation in hours worked. Wages are `S_BRUT` in the short panel and `SB` in the long panel. These are gross annual wages which include overtime and bonuses. We divide these by `NBHEUR`, the number of hours worked in the year, to get hourly wages. In the long panel, we use the CPI provided by INSEE to deflate all wages to 2015 euros. Our main short-panel results use the 2014–2015 panel, which needs no adjustment because consumer price inflation in 2015 was zero; the short panels for other years (Appendix D.5) are in nominal euros.

– Establishment: We use the INSEE provided establishment codes `SIRET` to define the boundaries of a team.

– Occupation: We also use 1-digit occupation codes, the first digit of `PCS4`, to define a team.

A.2 Quadratic Expansion

In this appendix, we derive computationally efficient formulas for the positive and negative components of the coworker effects in both the linear and quadratic cases, which we use to estimate Equations (4) and (5). These formulas can be computed in linear time after sorting the data, and do not require any look-aheads or look-backs through the data. We begin with the linear case. Suppose we have a sequence $\{x_i\}_{i=1}^N$ and we want to compute, for each $j \in \{1, 2, \ldots, N\}$, the leave-one-out average over the $N-1$ coworkers of j,

$$y_j^+ := \frac{1}{N-1} \sum_{i=1}^{N} \max\{x_i - x_j, 0\}$$

where the sum can run over all N workers because the own term $i = j$ contributes zero, so dividing by $N-1$ makes y_j^+ exactly the average gap to j's coworkers, matching the normalization of Equations (4) and (5). Without loss of generality, we will assume throughout that the sequence $\{x_i\}$ is sorted in ascending order, i.e. $x_1 \leq x_2 \leq x_3 \leq$

$\cdots \leq x_N$. Moreover, we use Z_j to denote the running sum

$$Z_j := \sum_{i=1}^{j} x_i$$

which can conveniently be computed in a single pass through the data (i.e., in linear time) since $Z_{j+1} = Z_j + x_{j+1}$. Using this, we can rewrite y_j^+ as

$$\begin{aligned} y_j^+ &= \frac{1}{N-1} \sum_{i=j+1}^{N} (x_i - x_j) && (x_i \geq x_j \text{ for } i \geq j) \\ &= \frac{1}{N-1} \left[\sum_{i=j+1}^{N} x_i - (N-j)x_j \right] \\ &= \frac{1}{N-1} \left[Z_N - Z_j - (N-j)x_j \right] \end{aligned}$$

We can also do the same for the negative component

$$\begin{aligned} y_j^- &:= \frac{1}{N-1} \sum_{i=1}^{N} \min\{x_i - x_j, 0\} \\ &= \frac{1}{N-1} \sum_{i=1}^{j-1} (x_i - x_j) && (x_i \geq x_j \text{ for } i \geq j) \\ &= \frac{1}{N-1} \left[Z_{j-1} - (j-1)x_j \right] \\ &= \frac{1}{N-1} \left[Z_j - x_j - (j-1)x_j \right] \\ &= \frac{1}{N-1} (Z_j - jx_j) \end{aligned}$$

We now turn to the quadratic terms and similarly define

$$Z_{2,j} := \sum_{i=1}^{j} x_i^2$$

to be the running sum of the x_i^2 terms up to j. The positive component in the quadratic case can then be rewritten as

$$\begin{aligned}
y_{2,j}^{+} &= \frac{1}{N-1}\sum_{i=1}^{N}\max\{x_i - x_j, 0\}^2 \\
&= \frac{1}{N-1}\sum_{i=j+1}^{N}(x_i - x_j)^2 \\
&= \frac{1}{N-1}\sum_{i=j+1}^{N}(x_i^2 - 2x_i x_j + x_j^2) \\
&= \frac{1}{N-1}\left[\sum_{i=j+1}^{N} x_i^2 - 2x_j\sum_{i=j+1}^{N} x_i + (N-j)x_j^2\right] \\
&= \frac{1}{N-1}\left[(Z_{2,N} - Z_{2,j}) - 2x_j(Z_N - Z_j) + (N-j)x_j^2\right]
\end{aligned}$$

Similarly, for the negative component, we have

$$\begin{aligned}
y_{2,j}^{-} &= \frac{1}{N-1}\sum_{i=1}^{N}\min\{x_i - x_j, 0\}^2 \\
&= \frac{1}{N-1}\sum_{i=1}^{j-1}(x_i - x_j)^2 \\
&= \frac{1}{N-1}\sum_{i=1}^{j-1}(x_i^2 - 2x_i x_j + x_j^2) \\
&= \frac{1}{N-1}\left[\sum_{i=1}^{j-1} x_i^2 - 2x_j\sum_{i=1}^{j-1} x_i + (j-1)x_j^2\right] \\
&= \frac{1}{N-1}\left[Z_{2,j-1} - 2x_j Z_{j-1} + (j-1)x_j^2\right] \\
&= \frac{1}{N-1}\left[(Z_{2,j} - x_j^2) - 2x_j(Z_j - x_j) + (j-1)x_j^2\right] \\
&= \frac{1}{N-1}\left[Z_{2,j} - 2x_j(Z_j - x_j) + (j-2)x_j^2\right]
\end{aligned}$$

B Identification: Bounds on the Ambient Gradient

This appendix states the assumptions under which the gradients in Tables 3 and 6 bound the ambient city-size component of annual wage growth.

B.1 Setup and notation

We posit that the wages of worker i at date t, employed last year at firm $j := j(i, t-1)$ and working this year in commuting zone $c := c(i, t)$ are governed by the following innovation equation

$$w_{i,t} \;=\; \nu\, w_{i,t-1} \;+\; A_{c(i,t)} \;+\; B_{j(i,t-1)} \;+\; \theta' z_{i,t} \;+\; X'_{i,t}\beta \;+\; \xi_i \;+\; u_{i,t}, \qquad \text{(M)}$$

where A_c is the ambient city component, B_j a firm component, $z_{i,t} \in \mathbb{R}^K$ the coworker controls ($K = 4$: the higher- and lower-wage gap averages and their squares), $X_{i,t}$ contains age and gender, and ξ_i is unobserved heterogeneity in growth potential. We maintain the standard assumption that

$$\mathrm{E}[\, u_{i,t} \mid w_{i,t-1},\; c,\; j,\; z_{i,t},\; X_{i,t},\; \xi_i] \;=\; 0.$$

Equation (M) describes the wage-growth innovation conditional on the current wage: among workers at the same $w_{i,t-1}$, wages next year differ systematically by commuting zone, firm, coworkers, demographics, and persistent individual drift. The first-difference specification of Appendix D.3 is the special case $\nu = 1$.

We then project our estimated city fixed effects down on log population. We define $\Pi[x_c]$ to be the slope of this projection for any random variable x_c defined at the commuting-zone level:

$$x_c \;=\; \text{const} \;+\; \Pi[x_c]\, p_c \;+\; \text{error}, \qquad \Pi[x_c] \;:=\; \frac{\mathrm{Cov}(x_c,\; p_c)}{\mathrm{Var}(p_c)}$$

Note that the operator Π is linear in x and $\Pi[\text{constant}] = 0$. Our object of interest is the ambient gradient

$$\gamma \;:=\; \Pi[A_c] \;=\; \frac{\mathrm{Cov}(A_c,\; p_c)}{\mathrm{Var}(p_c)}$$

We now define the four specifications S_k that we run in the main text as:

- specification 0 (baseline): $S_0 = (1, D_c, w_{i,t-1}, X_{i,t})$, with corresponding zone effects ψ_c^0
- specification 1 (firm effects): $S_1 = S_0 \cup D_j$, with corresponding zone effects ψ_c^1
- specification cw (coworkers only): $S_{\text{cw}} = S_0 \cup z_{i,t}$, with corresponding zone effects ψ_c^{cw}
- specification 2 (both): $S_2 = S_0 \cup D_j \cup z_{i,t}$, with corresponding zone effects ψ_c^2,

where D_c and D_j denote the full sets of zone and firm indicators. For each specification, the second stage regresses the estimated commuting zone coefficients ψ_c^k on log population, and the estimand of this appendix is its slope $\alpha^k := \Pi[\psi_c^k]$.

Finally, for any random variable W, define its component regressions as

$$\begin{aligned} W &= g_c^W + \ell^W + \varepsilon^W && \text{on } S_0, \\ W &= a_c^W + b_j^W + \ell^W + \varepsilon^W && \text{on } S_1, \\ W &= h_c^W + \ell^W + \varepsilon^W && \text{on } S_{\text{cw}}. \end{aligned}$$

where ℓ^W is fitted by the remaining controls and ε^W is orthogonal to the regressor set in that row.

B.2 Decomposition across specifications

In this section, we show that, under this data generating process, we can exactly decompose the city-size gradient into its ambient and sorting components for each specification that we report in the paper. We do so for the baseline specification, then for the specifications with firm fixed effects and with coworker controls, separately and together.

Lemma 1 (population decomposition)**.** *Under* (M) *and the maintained conditions,*

$$\alpha^0 = \gamma + \Pi[g_c^B] + \theta' \Pi[g_c^z] + \Pi[g_c^\xi].$$

Proof. The S_0 component regressions are

$$B_j = g_c^B + \ell^B + \varepsilon^B,$$
$$z_{i,t} = g_c^z + \ell^z + \varepsilon^z,$$
$$\xi_i = g_c^\xi + \ell^\xi + \varepsilon^\xi.$$

Substituting into (M) and collecting terms gives

$$w_{i,t} = \underbrace{A_c + g_c^B + \theta' g_c^z + g_c^\xi}_{\text{zone component}} + \underbrace{\nu w_{i,t-1} + X_{i,t}'\beta + \ell^B + \theta'\ell^z + \ell^\xi}_{\text{lagged wage and demographic controls}} + \underbrace{\varepsilon^B + \theta'\varepsilon^z + \varepsilon^\xi + u_{i,t}}_{\text{orthogonal to } S_0}.$$

Hence

$$\psi_c^0 = A_c + g_c^B + \theta' g_c^z + g_c^\xi + \text{constant}.$$

Applying Π yields the result. □

The baseline combines the ambient gradient with sorting on firm quality, coworker composition, and worker growth potential conditional on lagged wages and demographics. Note that this decomposition did not require us to impose any sign restrictions on the admissible sorting patterns.

In our setting, firm effects refer to lagged firm-occupation pairs. This means that the identification of separate commuting zone and firm effects would require a connected set linking firms to zones in which their workers are observed. With only single-zone firms, the effects cannot be separated, hence we use multi-zone firms.

Lemma 2 (firm effects)**.** *Under* (M) *and the maintained conditions, including connectedness,*

$$\alpha^1 = \gamma + \theta'\Pi[a_c^z] + \Pi[a_c^\xi].$$

Proof. The S_1 component regressions are

$$z_{i,t} = a_c^z + b_j^z + \ell^z + \varepsilon^z,$$
$$\xi_i = a_c^\xi + b_j^\xi + \ell^\xi + \varepsilon^\xi.$$

Substituting into (M) and collecting terms gives

$$\begin{aligned} w_{i,t} = &\underbrace{A_c + \theta' a_c^z + a_c^\xi}_{\text{zone component}} + \underbrace{B_j + \theta' b_j^z + b_j^\xi}_{\text{firm component}} \\ &+ \underbrace{\nu w_{i,t-1} + X_{i,t}'\beta + \theta'\ell^z + \ell^\xi}_{\text{lagged wage and demographic controls}} \\ &+ \underbrace{\theta'\varepsilon^z + \varepsilon^\xi + u_{i,t}}_{\text{orthogonal to } S_1}. \end{aligned}$$

Hence

$$\psi_c^1 = A_c + \theta' a_c^z + a_c^\xi + \text{constant},$$
$$\phi_j^1 = B_j + \theta' b_j^z + b_j^\xi + \text{constant}.$$

Applying the operator Π to the zone component yields the desired result. □

We next add coworker controls to the baseline without firm effects, leaving firm and worker sorting conditional on coworker composition.

Lemma 3 (coworkers without firm effects)**.** *Under* (M) *and the maintained conditions,*

$$\alpha^{\text{cw}} = \gamma + \Pi[h_c^B] + \Pi[h_c^\xi].$$

Proof. The S_{cw} component regressions are

$$B_j = h_c^B + \ell^B + \varepsilon^B,$$
$$\xi_i = h_c^\xi + \ell^\xi + \varepsilon^\xi.$$

Substituting into (M) and collecting terms gives

$$w_{i,t} = \underbrace{A_c + h_c^B + h_c^\xi}_{\text{zone component}} + \underbrace{\nu w_{i,t-1} + X_{i,t}'\beta + \theta' z_{i,t} + \ell^B + \ell^\xi}_{\text{lagged wage, demographic, and coworker controls}} + \underbrace{\varepsilon^B + \varepsilon^\xi + u_{i,t}}_{\text{orthogonal to } S_{\text{cw}}}.$$

Hence

$$\psi_c^{\text{cw}} = A_c + h_c^B + h_c^\xi + \text{constant}.$$

Applying Π yields the result. □

This specification does not rely on multi-zone firms to separate firm and zone effects, but coworker endogeneity can affect both conditional sorting terms.

Let $\tilde{z}$ be the residual from regressing $z_{i,t}$ on S_1, and define

$$\pi := \mathrm{E}[\tilde{z}\tilde{z}']^{-1}\mathrm{E}[\tilde{z}\xi_i].$$

The coworker–worker orthogonality condition is

$$\mathrm{E}[\tilde{z}\xi_i] = 0, \qquad \text{equivalently } \pi = 0. \tag{CWR$'$}$$

It rules out selection on growth potential in coworker composition after removing additive zone and firm effects, lagged wages, and demographics.

Lemma 4 (coworker controls)**.** *Under* (M) *and the maintained conditions, the coefficient on $z_{i,t}$ in Specification 2 is $\theta + \pi$, and*

$$\alpha^1 - \alpha^2 = (\theta + \pi)'\Pi[a_c^z],$$
$$\alpha^2 = \gamma - \pi'\Pi[a_c^z] + \Pi[a_c^\xi].$$

Under (CWR$'$), $\alpha^2 = \gamma + \Pi[a_c^\xi]$.

Proof. The S_1 component regressions are

$$z_{i,t} = a_c^z + b_j^z + \ell^z + \tilde{z}_{i,t},$$
$$\xi_i = a_c^\xi + b_j^\xi + \ell^\xi + \varepsilon^\xi.$$

By the definition of π, the residual

$$\varepsilon^{\xi,2} := \varepsilon^\xi - \pi' \tilde{z}_{i,t}$$

is orthogonal to S_2. Substituting $\varepsilon^\xi = \pi' \tilde{z}_{i,t} + \varepsilon^{\xi,2}$ and $\tilde{z}_{i,t} = z_{i,t} - a_c^z - b_j^z - \ell^z$ into (M) gives

$$\begin{aligned} w_{i,t} = & \underbrace{A_c + a_c^\xi - \pi' a_c^z}_{\text{zone component}} + \underbrace{B_j + b_j^\xi - \pi' b_j^z}_{\text{firm component}} \\ & + \underbrace{(\theta + \pi)' z_{i,t}}_{\text{coworker controls}} \\ & + \underbrace{\nu w_{i,t-1} + X_{i,t}' \beta + \ell^\xi - \pi' \ell^z}_{\text{lagged wage and demographic controls}} \\ & + \underbrace{\varepsilon^{\xi,2} + u_{i,t}}_{\text{orthogonal to } S_2}. \end{aligned}$$

Hence the coworker coefficient is $\theta + \pi$ and

$$\psi_c^2 = A_c + a_c^\xi - \pi' a_c^z + \text{constant}.$$

Applying Π and using Lemma 2 gives

$$\alpha^2 = \gamma + \Pi[a_c^\xi] - \pi' \Pi[a_c^z],$$
$$\alpha^1 - \alpha^2 = (\theta + \pi)' \Pi[a_c^z].$$

Under (CWR′), $\pi = 0$. □

When $\pi' \Pi[a_c^z] > 0$, coworker controls remove more of the city-size gradient than the structural coworker term explains. Attenuation can therefore arise even if $\theta = 0$.

B.3 Coefficient bounds

The bounds require the following restrictions on conditional sorting:

- **(SR-0′)** $\Pi[g_c^B] \geq 0$, $\theta'\Pi[g_c^z] \geq 0$, and $\Pi[g_c^\xi] \geq 0$: firm quality, coworker contributions, and worker growth potential each sort non-negatively with city size, conditional on lagged wages and demographics.
- **(SR-1′)** $\theta'\Pi[a_c^z] + \Pi[a_c^\xi] \geq 0$: within-firm coworker and worker sorting is non-negative on net.
- **(SR-cw′)** $\Pi[h_c^B] + \Pi[h_c^\xi] \geq 0$: firm and worker sorting conditional on coworker controls is non-negative on net.
- **(SR-2′)** $\Pi[a_c^\xi] \geq 0$: within-firm worker sorting is non-negative. Under (CWR′), this is also the worker zone component after adding coworker controls.

Theorem 1 (coefficient bounds)**.** *Under* (M) *and the maintained conditions:*

(i) (SR-0′) implies $\gamma \leq \alpha^0$

(ii) (SR-1′) implies $\gamma \leq \alpha^1$

(iii) (SR-cw′) implies $\gamma \leq \alpha^{\text{cw}}$

(iv) (CWR′) *and (SR-2′) imply* $\gamma \leq \alpha^2$

Proof. Apply the sign restrictions to Lemmas 1 to 4, respectively. □

These restrictions are not nested and, importantly, do not imply an ordering of the estimated gradients. If all four sign restrictions and (CWR′) hold, then

$$\gamma \leq \min\{\alpha^0, \alpha^1, \alpha^{\text{cw}}, \alpha^2\}.$$

These are individual upper bounds. They need not tighten as controls are added, because conditioning changes the remaining sorting components.

Ordering the bounds. Tightening requires non-negative city-size gradients in the contributions absorbed by the added controls. Let ϑ^{cw} be the coworker coefficient vector in S_{cw}, and ϕ_j^k the firm effect in S_k, $k \in \{1, 2\}$. The omitted-variable identities

give

$$\begin{aligned}
\alpha^0 - \alpha^1 &= \Pi[g_c^{\phi^1}], \\
\alpha^0 - \alpha^{\text{cw}} &= (\vartheta^{\text{cw}})'\Pi[g_c^z], \\
\alpha^1 - \alpha^2 &= (\theta + \pi)'\Pi[a_c^z], \\
\alpha^{\text{cw}} - \alpha^2 &= \Pi[h_c^{\phi^2}].
\end{aligned}$$

If these four contributions are non-negative, adding controls weakly lowers the gradient. Together with (CWR′) and (SR-2′), this gives

$$\gamma \leq \alpha^2 \leq \min\{\alpha^1, \alpha^{\text{cw}}\}, \qquad \max\{\alpha^1, \alpha^{\text{cw}}\} \leq \alpha^0.$$

The earlier sign restrictions do not imply these additional conditions: for example, (SR-1′) allows a negative $\theta'\Pi[a_c^z]$ offset by positive worker sorting. The bounds concern population coefficients and the ordering of their estimates is an empirical result, not a finite-sample implication of the assumptions.

B.4 The persistence coefficient

In this section, we discuss how to interpret changes in coefficients other than the key object of interest, the city-size gradient. In particular, the estimated lagged-wage coefficient changes from 0.91 at baseline to 0.81 with firm effects and 0.96 with coworker controls (Table 3). This raises two concerns: whether the decline in the city-size gradient merely reflects these changes in persistence, and whether estimating persistence in a short panel biases the city effects.

Changes across specifications. The decompositions allow the lagged-wage coefficient to change when controls are added. Writing λ^W for its coefficient in the baseline component regression of W, the same substitution used in Lemma 1 gives

$$\nu^0 = \nu + \lambda^B + \theta'\lambda^z + \lambda^\xi.$$

This equation implies that the baseline coefficient incorporates not only the true autoregressive component, but also to some extent the correlation between lagged wages and unobserved factors that are correlated with current wages. Thus the firm, coworker, and worker components affect both the estimated persistence coefficient and the zone effects. These are identities that govern the attenuation of the coefficients.

Crucially, they do not hold the persistence parameter ν^k fixed, which means the decomposition is robust to changes in ν^k (these changes would not introduce any additional correction terms into the bounds). However, the interpretation of those bounds still requires the sorting restrictions in Theorem 1.

The own-wage content of the coworker controls also helps explain why their inclusion can change the persistence coefficient. For the higher-wage gap,

$$z_{i,t}^{+} = \frac{1}{n_i - 1} \sum_{j \in \mathbb{W}_{i,t-1}^{+}} w_{j,t-1} - \frac{n_i^{+}}{n_i - 1} w_{i,t-1},$$

where n_i is team size and $\mathbb{W}_{i,t-1}^{+}$ contains its n_i^{+} higher-paid coworkers. A positive gap coefficient therefore includes a negative own-wage contribution. More generally, for fixed κ, replacing z by $z^* = z + \kappa w_{i,t-1}$ gives

$$\nu^k w_{i,t-1} + \vartheta' z = (\nu^k - \vartheta' \kappa) w_{i,t-1} + \vartheta' z^*,$$

where ϑ is the fitted coworker coefficient vector. This example shows that the reported lagged-wage coefficient can change without any change in fitted values or zone effects. We cannot apply this same reparameterization directly to the coworker gaps, since the share $n_i^{+}/(n_i - 1)$ varies across workers. But it helps explain why changes in ν^k, on their own, do not tell us whether the city-size gradient is mechanically distorted.

Short panels and first differences. One might also worry about Nickell bias, since we estimate a lagged-wage coefficient together with fixed effects using a short panel. Here, however, workers share the same fixed effects, so each effect is estimated using observations from multiple workers. Having only two observations per worker therefore does not create the same problem as in a short panel with worker fixed effects. This does not mean that our estimates are unbiased in finite samples—the identities above are population results. We examine this concern further in Appendix D.6, where we report results using longer panels and worker fixed effects.

Estimating the model in first differences without controlling for lagged wages would impose $\nu = 1$. There is no reason for this restriction to resolve the problem by itself. If $\nu \neq 1$, the wage-growth regression omits $(\nu - 1) w_{i,t-1}$, and the zone and coworker controls may pick up part of this omitted term. In the log-wage results, the lower-

paid-coworker coefficient rises from -0.14 ($t = -10.9$) in levels to 0.10 ($t = 10.6$) in first differences. We therefore estimate persistence freely in the main specification and report first differences separately (Appendix D.3).

B.5 The empirical bounds

Table B.1 collects the second-stage gradients from Tables 3 and 6 with the restriction each bound invokes.

Table B.1: Coefficient Bounds on the Ambient Gradient

Specification	Full sample	Stayers	Restriction	Bound on γ
Baseline	0.136	0.078	(SR-0′)	$\gamma \leq 0.136$
	(0.016)	(0.013)		
Firm fixed effects	0.070	0.030	(SR-1′)	$\gamma \leq 0.070$
	(0.016)	(0.008)		
Coworker controls	0.073	0.049	(SR-cw′)	$\gamma \leq 0.073$
	(0.011)	(0.012)		
Both	0.058	0.022	(CWR′) + (SR-2′)	$\gamma \leq 0.058$
	(0.014)	(0.008)		

Note: Hourly wages in euros; the full-sample and stayer columns reproduce Panel B of Tables 3 and 6 (robust standard errors in parentheses; 304 commuting zones). In each row, we derive the bound from Theorem 1 under the corresponding restriction, both for our full sample and for job stayers.

The long panel provides a separate check: across the five wage measures in Appendix D.6, adding controls reduces the baseline gradient by 47–92%, and the fully controlled stayer gradient is statistically indistinguishable from zero in four of the five measures.

C Relationship to De la Roca and Puga (2017)

In this appendix, we show that our specification in Equation (1) restricted to job stayers nests the dynamic specification from De la Roca and Puga (2017) in first differences. To begin, let us start with their dynamic specification (Equation 1, Section 3, p. 113). They posit a model of wages where

$$w_{it} = \sigma_{c(i,t)} + \mu_i + \sum_{j=1}^{C} \delta_{j,c(i,t)} e_{ijt} + \mathbf{x}_{it}'\beta + \varepsilon_{it} \tag{C.1}$$

If we take first differences:

$$\Delta w_{it} = \sigma_{c(i,t)} - \sigma_{c(i,t-1)} + \left[\sum_{j=1}^{C} \delta_{j,c(i,t)} e_{ijt} - \sum_{j=1}^{C} \delta_{j,c(i,t-1)} e_{ij,t-1} \right] + \Delta \mathbf{x}_{it}'\beta + \Delta\varepsilon_{it} \tag{C.2}$$

In our stayers sample, we know that, by construction $c(i,t) = c(i,t-1)$ so $\sigma_{c(i,t)} - \sigma_{c(i,t-1)} = 0$. Moreover, we also know that $\delta_{j,c(i,t)} = \delta_{j,c(i,t-1)}$. So, for our stayers sample we can simplify Equation (C.2) to

$$\Delta w_{it} = \sum_{j=1}^{C} \delta_{j,c(i,t)} \Delta e_{ijt} + \Delta \mathbf{x}_{it}'\beta + \Delta\varepsilon_{it} \tag{C.3}$$

But, the structure of e_{ijt}, which is the cumulative number of years that worker i has spent in city j up until time t means that $\Delta e_{ijt} = e_{ijt} - e_{ij,t-1} = \mathbb{1}\{j = c(i,t)\}$. This means that the terms in the summand drop out for all $j \neq c(i,t)$, and we obtain

$$\Delta w_{it} = \delta_{c(i,t),c(i,t)} + \Delta \mathbf{x}_{it}'\beta + \Delta\varepsilon_{it} \tag{C.4}$$

So, when we estimate:

$$w_{i,t} = \nu w_{i,t-1} + \psi_{c(i,t)} + \epsilon_{i,t} \tag{1}$$

in our sample of job stayers, our commuting zone fixed effects $\psi_{c(i,t)}$ sweep out the δ terms, and our inclusion of the lagged wage $w_{i,t-1}$ nests Equation (C.4) in the case where $\nu = 1$.[12] In our stayers regressions, the coefficients that De la Roca and Puga

[12] Note: many of the worker level controls that De la Roca and Puga (2017) include in $\mathbf{x}_{it}$ are also largely swept out when taking first differences (for instance, the linear components of experience and job tenure). In our specifications we also add fixed effects for observable worker characteristics such as gender and age.

(2017) identify as the dynamic effects that accrue to experience in each city type are captured in the commuting zone fixed effects $\psi_{c(i,t)}$.

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D Results Appendix

D.1 Robustness to Log Wages

Table D.1: Decomposing the Urban Wage Premium: Log Hourly Wages

	(1)	(2)	(3)	(4)
	Baseline	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions				
Lagged Wage (ν)	0.8897	0.6702	0.9164	0.7412
	(0.0012)	(0.0081)	(0.0018)	(0.0164)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)			0.2791	0.1338
			(0.0054)	(0.0124)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)			-0.0108	-0.1401
			(0.0081)	(0.0128)
Nonlinear Effects ($\tilde{\theta}_2^+$)			0.0838	0.0767
			(0.0087)	(0.0055)
Nonlinear Effects ($\tilde{\theta}_2^-$)			-0.0447	-0.0579
			(0.0121)	(0.0198)
Lagged Firm Fixed Effects	No	Yes	No	Yes
Observations	16,628,524	14,953,462	14,698,578	14,698,578
R^2	0.8245	0.8508	0.8238	0.8540
Panel B: Second Stage Regressions				
Log Population (α)	0.00573	0.00329	0.00326	0.00244
	(0.0006)	(0.0007)	(0.0004)	(0.0006)
Observations	304	304	304	304
R^2	0.3321	0.0980	0.2109	0.0689
$\hat{\alpha}$ change relative to (1)	–	-42.6%	-43.0%	-57.4%

Note: Panel A reports first-stage regressions where the dependent variable is log hourly wages. Column (1) shows the baseline specification from Equation (1) with commuting zone fixed effects and lagged wages. Column (2) adds firm fixed effects following Equation (3). Column (3) includes the full coworker effects specification from Equation (4). Column (4) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust. Sample consists of all workers in the short panel with residualized wages for age and gender.

Table D.2: The Urban Wage Premium Holding Job Mobility Constant: Log Hourly Wages

	(1)	(2)	(3)	(4)	(5)
	Baseline (Full Sample)	Baseline (Stayers)	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions					
Lagged Wage (ν)	0.8897	0.9642	0.8233	0.9688	0.8612
	(0.0012)	(0.0034)	(0.0133)	(0.0030)	(0.0133)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)				0.2560	0.2028
				(0.0114)	(0.0088)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)				-0.0427	-0.1630
				(0.0131)	(0.0130)
Nonlinear Effects ($\tilde{\theta}_2^+$)				0.0273	0.0534
				(0.0112)	(0.0091)
Nonlinear Effects ($\tilde{\theta}_2^-$)				-0.0385	-0.0447
				(0.0157)	(0.0168)
Lagged Firm Fixed Effects	No	No	Yes	No	Yes
Observations	16,628,524	9,877,713	8,779,943	8,844,662	8,644,021
R^2	0.8245	0.9268	0.9427	0.9273	0.9449
Panel B: Second Stage Regressions					
Log Population (α)	0.00573	0.00262	0.00142	0.00145	0.00058
	(0.0006)	(0.0003)	(0.0004)	(0.0003)	(0.0003)
Observations	304	304	304	304	304
R^2	0.3321	0.1586	0.0613	0.0496	0.0134
$\hat{\alpha}$ change relative to (1)	–	-54.3%	-75.2%	-74.6%	-89.9%

Note: Panel A reports first-stage regressions where the dependent variable is log hourly wages. Column (1) shows the baseline specification from Equation (1) for the full sample. Columns (2)-(5) restrict to job stayers. Column (2) shows the baseline specification from Equation (1) restricting the sample to job stayers. Column (3) adds firm fixed effects following Equation (3). Column (4) includes the full coworker effects specification from Equation (4). Column (5) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust. Wages are residualized for age and gender.

Table D.3: Job-to-Job Transition Rates and City Size: Log Hourly Wages

	(1)	(2)
Dependent variable	EE Transition	EE Transition
Log Population	0.0294	0.0469
	(0.0018)	(0.0030)
Lagged Log Wage		-0.1047
		(0.0016)
Controls	No	Lagged Wage
Fixed Effects	No	No
SE Clustering	CZ	CZ
Observations	16,628,524	16,628,524
R^2	0.0022	0.0155

Note: This table reports regressions of job-to-job (EE) transition rates on city size. Column (1) shows the baseline specification from Equation (6). Column (2) adds lagged log hourly wages as a control following Equation (7). Standard errors are clustered at the commuting zone (CZ) level. The sample consists of all workers from one cross-section of the short panel.

D.2 Robustness to Trimming Extreme Wages

Our euro-denominated wage measure has a heavy right tail, even after we account for recording errors (as described in Section 1). In this appendix we winsorize hourly wages and lagged hourly wages at their 0.1st and 99.9th percentiles (for 2015, €2.71 and €150) and repeat the full decomposition. We find that our main results are robust to this additional data cleaning measure. Winsorizing moves the baseline premium from 0.136 to 0.106, and the premium with our preferred control set from 0.058 to 0.053. That is, the attenuation in the winsorized sample is 50.2% rather than 57.3% in the untrimmed sample. Moreover, when we examine the Gelbach decomposition of the attenuation, we see that the results are remarkably similar to our main specification. If anything, they would attribute a slightly larger share of the attenuation to the inclusion of coworker controls. We conclude that neither the attenuation nor its firm/coworker attribution is driven by the remaining tail.

Table D.4: Decomposing the Urban Wage Premium: Winsorized Wages

	(1)	(2)	(3)	(4)
	Baseline	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions				
Lagged Wage (ν)	0.9463	0.8358	0.9732	0.8856
	(0.0096)	(0.0236)	(0.0053)	(0.0274)
Coworker Controls	No	No	Yes	Yes
Lagged Firm Fixed Effects	No	Yes	No	Yes
Observations	16,628,524	14,953,462	14,698,578	14,698,578
Panel B: Second Stage Regressions				
Log Population (α)	0.10582	0.06605	0.05863	0.05275
	(0.0129)	(0.0143)	(0.0098)	(0.0125)
Observations	304	304	304	304
$\hat{\alpha}$ change relative to (1)	–	-37.6%	-44.6%	-50.2%

Note: Same specifications as Table 3, with hourly wages and lagged hourly wages winsorized at their 0.1st and 99.9th percentiles before estimation. Panel A reports the first-stage persistence coefficient; the coworker-control coefficients are omitted for space. Panel B reports the projection of estimated commuting zone fixed effects onto logged commuting zone population. First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust.

Table D.5: Job Stayers with Winsorized Wages

	(1)	(2)	(3)	(4)	(5)
	Baseline (Full Sample)	Baseline (Stayers)	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions					
Lagged Wage (ν)	0.9463	1.0048	0.9529	0.9946	0.9201
	(0.0096)	(0.0093)	(0.0180)	(0.0078)	(0.0242)
Coworker Controls	No	No	No	Yes	Yes
Lagged Firm Fixed Effects	No	No	Yes	No	Yes
Observations	16,628,524	9,877,713	8,779,943	8,844,662	8,644,021
Panel B: Second Stage Regressions					
Log Population (α)	0.10582	0.05077	0.02395	0.03039	0.01390
	(0.0129)	(0.0108)	(0.0072)	(0.0113)	(0.0058)
Observations	304	304	304	304	304
$\hat{\alpha}$ change relative to (1)	–	-52.0%	-77.4%	-71.3%	-86.9%

Note: Same specifications as Table 6, with hourly wages and lagged hourly wages winsorized at their 0.1st and 99.9th percentiles before estimation. Column (1) is the winsorized full-sample baseline; columns (2)-(5) restrict to job stayers. The coworker-control coefficients are omitted for space. First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust.

D.3 Robustness to First-Differenced Wages

In this appendix we repeat the decomposition in first differences, replacing the freely estimated persistence coefficient with the restriction $\nu = 1$, so the dependent variable is the one-year change in hourly wages, $\Delta w_{i,t} = w_{i,t} - w_{i,t-1}$. This is the specification of Equation (1').

Table D.6: Decomposing the Urban Wage Premium: First-Differenced Wages

	(1)	(2)	(3)	(4)
	Baseline	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions				
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)			0.1909	0.1999
			(0.0111)	(0.0311)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)			0.0776	0.0707
			(0.0477)	(0.0527)
Nonlinear Effects ($\tilde{\theta}_2^+$)			-0.0003	0.0001
			(0.0000)	(0.0004)
Nonlinear Effects ($\tilde{\theta}_2^-$)			-0.0005	-0.0004
			(0.0001)	(0.0000)
Lagged Firm Fixed Effects	No	Yes	No	Yes
Observations	16,628,524	14,953,462	14,698,578	14,698,578
R^2	0.0046	0.1427	0.0526	0.1877
Panel B: Second Stage Regressions				
Log Population (α)	0.06444	0.04293	0.05505	0.04301
	(0.0105)	(0.0110)	(0.0104)	(0.0107)
Observations	304	304	304	304
R^2	0.1237	0.0629	0.0819	0.0594
$\hat{\alpha}$ change relative to (1)	–	-33.4%	-14.6%	-33.3%

Note: Panel A reports first-stage regressions where the dependent variable is the one-year first difference of hourly wages ($\Delta w_{i,t} = w_{i,t} - w_{i,t-1}$), the $\nu = 1$ restriction of the main specification; no lagged-wage regressor appears. Column (1) shows the baseline specification from Equation (1') with commuting zone fixed effects. Similarly, columns (2)-(4) report the results from the specifications where we change Equations (3) to (5) to use first-differenced wages as the dependent variable. Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust. Sample consists of all workers in the short panel with residualized wages for age and gender.

Table D.7: Job Stayers with First-Differenced Wages

	(1)	(2)	(3)	(4)	(5)
	Baseline (Full Sample)	Baseline (Stayers)	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions					
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)				0.1514	0.1789
				(0.0143)	(0.0248)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)				-0.0416	-0.0791
				(0.0368)	(0.0318)
Nonlinear Effects ($\tilde{\theta}_2^+$)				-0.0002	0.0009
				(0.0000)	(0.0006)
Nonlinear Effects ($\tilde{\theta}_2^-$)				-0.0004	-0.0004
				(0.0001)	(0.0000)
Lagged Firm Fixed Effects	No	No	Yes	No	Yes
Observations	16,628,524	9,877,713	8,779,943	8,844,662	8,644,021
R^2	0.0046	0.0032	0.2338	0.0312	0.2653
Panel B: Second Stage Regressions					
Log Population (α)	0.06444	0.05630	0.01562	0.02786	0.00007
	(0.0105)	(0.0119)	(0.0066)	(0.0120)	(0.0068)
Observations	304	304	304	304	304
R^2	0.1237	0.0479	0.0147	0.0092	0.0000
$\hat{\alpha}$ change relative to (1)	–	-12.6%	-75.8%	-56.8%	-99.9%

Note: Panel A reports first-stage regressions where the dependent variable is the one-year first difference of hourly wages ($\Delta w_{i,t} = w_{i,t} - w_{i,t-1}$); no lagged-wage regressor appears. Column (1) shows the baseline specification from Equation (1') with commuting zone fixed effects. Columns (2)-(5) restrict to job stayers. Column (2) shows the baseline specification from Equation (1') restricting the sample to job stayers. Similarly, columns (3)-(5) report the results from the specifications where we change Equations (3) to (5) to use first-differenced wages as the dependent variable. Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust. Wages are residualized for age and gender.

D.4 Other Moments

Table D.8: Adding Moments of the Coworker Wage Distribution

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Baseline	+Mean	+Variance	+Skewness	+Kurtosis	+P1	+P10	+P90	+P99
Panel A: First Stage Regressions									
Lagged Wage	0.9140	0.9604	0.9472	0.9101	0.9004	0.8382	0.7886	0.6972	0.6962
	(0.0203)	(0.0135)	(0.0162)	(0.0083)	(0.0099)	(0.0129)	(0.0134)	(0.0107)	(0.0096)
Mean Wage		0.0631	0.0484	0.0547	0.0569	-0.0027	-0.1300	-0.2341	-0.2310
		(0.0104)	(0.0230)	(0.0205)	(0.0198)	(0.0154)	(0.0142)	(0.0223)	(0.0277)
Mean Wage × Lagged Wage		-0.0012	-0.0014	-0.0014	-0.0014	-0.0016	-0.0013	-0.0013	-0.0014
		(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0004)	(0.0004)
Variance			0.0017	0.0017	0.0016	0.0017	0.0017	0.0000	-0.0002
			(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Variance × Lagged Wage			0.0000	0.0000	0.0000	0.0000	0.0000	-0.0000	-0.0000
			(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Skewness				-0.1548	-0.1473	-0.4751	-0.5294	-0.6525	-0.2529
				(0.0788)	(0.1366)	(0.1527)	(0.1743)	(0.2038)	(0.1101)
Skewness × Lagged Wage				0.0115	0.0175	0.0336	0.0315	0.0365	0.0127
				(0.0031)	(0.0071)	(0.0078)	(0.0082)	(0.0099)	(0.0083)
Kurtosis					-0.0024	0.0140	0.0110	0.0124	-0.0050
					(0.0051)	(0.0050)	(0.0054)	(0.0060)	(0.0034)
Kurtosis × Lagged Wage					-0.0003	-0.0010	-0.0009	-0.0008	0.0001
					(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
P1						0.2194	0.0101	0.0040	-0.0384
						(0.0134)	(0.0162)	(0.0129)	(0.0311)
P1 × Lagged Wage						0.0015	-0.0029	-0.0016	0.0002
						(0.0003)	(0.0009)	(0.0008)	(0.0009)
P10							0.4387	0.3503	0.3874
							(0.0231)	(0.0180)	(0.0305)
P10 × Lagged Wage							0.0039	0.0015	0.0002
							(0.0010)	(0.0009)	(0.0010)
P90								0.1656	0.1881
								(0.0171)	(0.0081)
P90 × Lagged Wage								0.0009	0.0000
								(0.0001)	(0.0001)
P99									-0.0102
									(0.0096)
P99 × Lagged Wage									0.0007
									(0.0001)
Observations	16,628,524	14,698,578	14,311,562	14,299,943	14,299,943	14,299,943	14,299,943	14,299,943	14,299,943
R^2	0.7177	0.7119	0.7322	0.7331	0.7334	0.7439	0.7539	0.7702	0.7734
Panel B: Second Stage Regressions									
Log Population (α)	0.13579	0.09627	0.10733	0.10554	0.10547	0.11668	0.07823	0.05445	0.05639
	(0.0164)	(0.0145)	(0.0129)	(0.0120)	(0.0120)	(0.0132)	(0.0104)	(0.0056)	(0.0054)
$\hat{\alpha}$ change relative to (1)	–	-29.1%	-21.0%	-22.3%	-22.3%	-14.1%	-42.4%	-59.9%	-58.5%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wages. All specifications include commuting zone fixed effects, age, and gender controls. Column (1) includes lagged wages. Column (2) adds mean wage and its interaction with lagged wage. Column (3) adds variance and its interaction with lagged wage. Column (4) adds skewness and its interaction. Column (5) adds kurtosis and its interaction. Column (6) adds the 1st percentile (P1) and its interaction. Column (7) adds the 10th percentile (P10) and its interaction. Column (8) adds the 90th percentile (P90) and its interaction. Column (9) adds the 99th percentile (P99) and its interaction. Panel B reports the projection of estimated commuting zone fixed effects from the first stage onto logged commuting zone population. First-stage standard errors are clustered at the commuting zone level. Second-stage standard errors are heteroskedasticity-robust. Sample consists of all workers in the short panel with residualized wages for age and gender.

D.5 Results for Other Years

Our main results use the 2014–2015 short panel. In this appendix we repeat the main decomposition in the 2012–2013, 2015–2016, and 2016–2017 short panels, in both euro levels and logs, and we additionally report the baseline gradient for the 2013–

2014 panel. The attenuation, if anything, is stronger in other years. The 2016 data contain unusually many extreme recorded wages; even after trimming, the euro-level estimates remain atypical, with a larger baseline gradient, lower persistence, and a negative coworker contribution. The log-wage results and fully controlled stayer gradients are comparable to other years, suggesting that the anomaly is concentrated in the euro-level estimates.

Table D.9: The Main Decomposition in Other Years

	2013	2014	2015	2016	2017
Panel A: Hourly Wages (€)					
Baseline (α^0)	0.1468	0.1325	0.1358	0.4592	0.1260
	(0.0173)	(0.0189)	(0.0164)	(0.0479)	(0.0148)
Coworkers + Firm FE (α^2)	0.0578	–	0.0580	0.0925	0.0525
	(0.0143)	–	(0.0140)	(0.0249)	(0.0103)
$\hat{\alpha}$ change	-60.6%	–	-57.3%	-79.9%	-58.3%
Stayers: Baseline (α_s^0)	0.0756	–	0.0782	0.3534	0.0513
	(0.0126)	–	(0.0130)	(0.0361)	(0.0114)
Stayers: Coworkers + Firm FE (α_s^2)	0.0087	–	0.0222	0.0200	0.0164
	(0.0091)	–	(0.0078)	(0.0157)	(0.0086)
$\hat{\alpha}$ change, stayers	-94.1%	–	-83.7%	-95.6%	-87.0%
Gelbach: firm channel (δ_{firm})	0.060	–	0.031	0.551	0.018
Gelbach: coworker channel (δ_{cw})	0.029	–	0.047	-0.185	0.055
Panel B: Log Hourly Wages					
Baseline (α^0)	0.0058	0.0060	0.0057	0.0083	0.0065
	(0.0007)	(0.0008)	(0.0006)	(0.0008)	(0.0006)
Coworkers + Firm FE (α^2)	0.0027	–	0.0024	0.0023	0.0025
	(0.0007)	–	(0.0006)	(0.0007)	(0.0005)
$\hat{\alpha}$ change	-52.9%	–	-57.4%	-72.7%	-61.8%
Stayers: Baseline (α_s^0)	0.0025	–	0.0026	0.0047	0.0030
	(0.0005)	–	(0.0003)	(0.0005)	(0.0005)
Stayers: Coworkers + Firm FE (α_s^2)	0.0009	–	0.0006	0.0003	0.0011
	(0.0004)	–	(0.0003)	(0.0005)	(0.0004)
$\hat{\alpha}$ change, stayers	-84.9%	–	-89.9%	-96.0%	-82.9%
Gelbach: firm channel (δ_{firm})	0.0018	–	0.0020	0.0043	0.0019
Gelbach: coworker channel (δ_{cw})	0.0013	–	0.0013	0.0018	0.0021
Commuting zones	304	304	304	304	304

Note: Each column repeats the specifications of Table 3 in a different two-year short panel, labeled by its second (outcome) year; the 2015 column reproduces the main tables. α^0 and α^2 are the second-stage projections of the baseline and fully controlled commuting zone fixed effects on log population; the Gelbach rows decompose $\alpha^0 - \alpha^2$ into the parts absorbed by the firm and coworker controls. Robust standard errors in parentheses. Euro magnitudes are nominal and not comparable across years. The stayer rows repeat the baseline and fully controlled specifications of Table 6 within each year's job-stayer subsample; the stayer $\hat{\alpha}$ change is computed relative to the full-sample baseline α^0. The 2014 source files carry no lagged employer identifiers, hence why only the baseline is available.

D.6 Evidence from the Long Panel

Our main results use two-year cross-sections of the near-universe short panel. In this appendix we ask whether the same decomposition holds within workers' lifetime employment histories, using the 1-in-12 long panel described in Section 1 (2003–2019, with the recording-error cleaning described there). We estimate the same first-stage regressions, hourly wages on the lagged wage, commuting zone effects, age, and gender, adding lagged firm fixed effects and coworker controls in stages, and project the estimated zone effects on log population. We run those regressions for five wage measures: the three from the cross-section (euro levels, logs, ranks) and two annual-earnings measures (earnings and log earnings), which carry no hours denominator, so that hours mismeasurement cannot drive their results. We report four exercises: the decomposition (Table D.10), the job-stayer subsample (Table D.11), a worker-fixed-effects variant (Table D.12), and job-to-job transition rates (Table D.13). The results are broadly consistent with the short-panel findings in the main text.

Table D.10: Long Panel: AR(1) Decomposition, Five Wage Measures

	Hourly (€)	Log Hourly	Rank	Earnings (€)	Log Earnings
Panel A: First Stage Persistence (ν)					
Baseline	0.9554	0.9185	0.8623	0.9353	0.6777
Coworkers + Firm FE	0.8165	0.7069	0.6478	0.7195	0.3854
Panel B: Second Stage (α, on log population)					
Baseline (α^0)	0.0655	0.0039	0.2329	176.9	0.0161
	(0.0099)	(0.0005)	(0.0220)	(25.3)	(0.0016)
+ Lagged Firm FE	0.0511	0.0028	0.1468	81.9	0.0048
	(0.0110)	(0.0007)	(0.0297)	(24.7)	(0.0014)
+ Coworkers	0.0455	0.0039	0.2281	155.4	0.0144
	(0.0083)	(0.0005)	(0.0219)	(25.0)	(0.0015)
Coworkers + Firm FE (α^2)	0.0222	0.0019	0.1244	13.4	0.0027
	(0.0128)	(0.0008)	(0.0337)	(24.5)	(0.0014)
$\hat{\alpha}$ change relative to baseline	-66.1%	-52.6%	-46.6%	-92.5%	-82.9%
Gelbach: firm channel (δ_{firm})	0.006	0.0011	0.095	19.1	0.0101
Gelbach: coworker channel (δ_{cw})	0.038	0.0010	0.013	144.4	0.0033

Note: Long panel, 2003–2019, cleaned as described in Section 1. Our first-stage specifications mirror Table 3. We regress the wage measure on its own lag and commuting zone, year, age and gender effects. We also add lagged firm fixed effects and coworker controls, both separately and together. The baseline specification has 13,284,705 observations, and in the specification with all of the controls we have 10,727,522. Panel B reports the projection of the commuting zone effects in the first stage on the natural log of population. The Gelbach rows show how much of the decline from α^0 to α^2 is absorbed by the firm and coworker channels.

Table D.11: Long Panel: Job Stayers

	Hourly (€)	Log Hourly	Rank	Earnings (€)	Log Earnings
Baseline, stayers (α)	0.0453	0.0027	0.1809	190.6	0.0173
	(0.0078)	(0.0004)	(0.0179)	(29.0)	(0.0017)
Coworkers + Firm FE, stayers (α)	-0.0099	0.0002	0.0398	4.7	0.0039
	(0.0111)	(0.0007)	(0.0327)	(32.6)	(0.0017)

Note: Same design as Table D.10, restricted to job stayers (no change of establishment × 1-digit occupation between $t-1$ and t, the definition of the cross-section tables); 10,191,749 observations in the stayer baseline and 8,811,051 with full controls. The fully controlled stayer gradient is statistically indistinguishable from zero in every measure except log earnings ($t = 2.3$).

Table D.12: Long Panel: Adding Worker Fixed Effects

	Hourly (€)	Log Hourly	Rank	Earnings (€)	Log Earnings
Baseline + Worker FE (α)	0.0897	0.0053	0.3212	119.1	0.0047
	(0.0211)	(0.0011)	(0.0557)	(42.7)	(0.0021)
Coworkers + Firm FE + Worker FE (α)	0.0672	0.0028	0.1637	103.9	0.0010
	(0.0271)	(0.0014)	(0.0647)	(56.8)	(0.0031)

Note: This table repeats the specifications in Table D.10, adding worker fixed effects to each regression. There are 13,018,276 observations in the baseline and 10,489,382 with all controls. With worker fixed effects, the commuting zone effects are identified from workers observed in more than one commuting zone over 2003–2019. The lagged-wage coefficient is then subject to the usual downward dynamic-panel bias, and falls to 0.58 in euro levels and 0.50 in logs. We therefore do not use these estimates to bound the ambient component, which rests on the short-panel specifications and Appendix B. Our interest here is only in whether accounting more fully for worker heterogeneity reverses the decomposition, and indeed, it does not.

Table D.13: Long Panel: Job-to-Job Transition Rates and City Size

	(1)	(2)	(3)	(4)
Log Population	0.0185	0.0200	0.0182	0.0173
	(0.0009)	(0.0015)	(0.0010)	(0.0014)
Lagged Log Wage	No	Yes	No	Yes
Year, Age, Sex FE	No	No	Yes	Yes
Observations	12,433,945	12,433,945	12,433,945	12,433,945

Note: Linear probability models of a job-to-job transition on log commuting zone population in the long panel, 2004–2019, standard errors clustered at the commuting zone level. In each year, we observe that the gradient is positive and significant, ranging from 0.015 to 0.022, and corroborating Table 5 on an independent sample.

Table A.1: Occupation Coding in DADS Data

1		**Farmers**
2		**Craftsmen, Tradespeople, and Business Owners**
3		**Executives and High-Level Professionals**
	31	*Independent Professionals*
	311c	Dentists
	311d	Psychologists and Therapists
	311e	Veterinarians
	3121	Lawyers
	33	*Public Sector Executives*
	34	*Professors, Scientific Professionals*
	342b	Research Professors
	344a	Hospital Doctors Without an Independent Practice
	344c	Residents in Medicine, Dentistry and Pharmacy
	344d	Salaried Pharmacists
	35	*Careers in Media, Arts, and Entertainment*
	352a	Journalists
	352b	Literary Authors, Screenwriters
	353a	Newspaper Editors, Media Executives, Publishing Directors
	37	*Corporate Administrative and Commercial Managers*
	372e	Legal Professionals
	375a	Advertising Executives
	376a	Financial Market Executives
	38	*Engineers and Technical Managers*
	382b	Salaried Architects
	384b	Mechanical Manufacturing Engineers and Metalworking Managers
	387c	Production Process Engineers and Managers
	387d	Quality Control Engineers and Managers
4		**Intermediate Professions**
5		**Clerical Workers**
6		**Manual Laborers**
9		**Non-Coded**

Note: This table illustrates occupation coding in the DADS. There are 6 distinct 1-digit occupation codes. To illustrate the granularity of 2-digit occupation coding, we also report a selection of the 2-digit and 4-digit codes for a representative example. The full list of occupation codes can be found on INSEE's website at https://www.insee.fr/fr/information/2497952.